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On the deformation of rings and algebras: III

By MURRAY GERSTENHABER*

This paper exhibits ways in which an algebra can be deformed. In Chapter I it is shown, in particular, that if an algebra over a field has an obstructed derivation, then it has a non-trivial deformation. Chapter II develops explicit formulas which can be used to define a new associative multiplication on the underlying module of an algebra. These are applied in Chapter III; first, to show that for every prime p the ring $\mathbf{Z}[x_0, \cdots, x_n]$ has a canonical deformation to a non-commutative ring; and second, to construct finite dimensional associative division algebras by deforming inseparable fields.

Chapter I begins with the definition of abstract "composition complexes", special cases of which are

(i) the cochain complex $C^*(X, k)$ of a finite simplicial complex X with coefficients in a commutative associative ring k with unit,

(ii) the cochain complex $C^*(A, A)$ of a k-algebra A with coefficients in itself, and

(iii) the tensor algebra over a Hopf k-algebra.

Composition complexes have natural cohomology and deformation theories, and we use them systematically through Chapter II; first, because it simplifies certain proofs; and second, because the general results have interesting specific interpretations in the three cases mentioned.

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CHAPTER I. DEFORMATION THEORY OF COMPOSITION COMPLEXES

1. Composition complexes

Let k be a commutative associative ring with unit. A *composition complex* with coefficients in k is a sequence $\mathcal{C} = \{C^0, C^1, \cdots\}$ of unital k-modules with the following properties:

(a) For every $f^m \in C^m$, $g^n \in C^n$ and i with $0 \leqq i \leqq m-1$, there is defined an element $f^m \circ_i g^n \in C^{m+n-1}$, bilinear in f and g, such that

$$(f^m \circ_i g^n) \circ_j h^l = \begin{cases} (f^m \circ_j h^l) \circ_{i+l-1} g^n & \text{if} \quad 0 \leqq j \leqq i-1 \\ f^m \circ_i (g^n \circ_{j-i} h^l) & \text{if} \quad i \leqq j \leqq n-1 . \end{cases}$$

(b) For every f^m and g^n, there is defined an element $f^m \smile g^n \in C^{m+n}$, bilinear in f and g, such that

$$(f^m \smile g^n) \circ_i h = \begin{cases} (f^m \circ_i h) \smile g^n & \text{if} \quad 0 \leqq i \leqq m-1 \\ f^m \smile (g^n \circ_{i-m} h) & \text{if} \quad m \leqq i \leqq m+n-1 . \end{cases}$$

Examples. (1) Let X be a finite, ordered simplicial complex, set $C^m = C^m(X, k)$; if $a_0 < \cdots < a_{m+n-1}$ are vertices, in order, spanning an $m+n-1$ simplex, then set

$$(f^m \circ_i g^n)(a_0, \cdots, a_{m+n-1}) = f^m(a_0, \cdots, a_i, a_{i+n}, \cdots, a_{m+n-1}) g^n(a_i, \cdots, a_{i+n}) ,$$

and if $a_0 < \cdots < a_{m+n}$ span a simplex, then set

$$(f^m \smile g^n)(a_0, \cdots, a_{m+n}) = f^m(a_0, \cdots, a_m) g^n(a_m, \cdots, a_{m+n}) .$$

If $m = 0$, set

$$f^m \circ_i g^n = 0 .$$

(2) Let A be a k-algebra, associative or not, set $C^m = C^m(A, A)$ (the set of k-multilinear maps from $A \times A \cdots \times A$ (n times) to A), set

$$(f^m \circ_i g^n)(a_1, \cdots, a_{m+n-1}) = f^m\big(a_1, \cdots, a_i, g^n(a_{i+1}, \cdots, a_{i+n-1}), a_{i+n}, \cdots, a_{m+n-1}\big) ,$$

and set

$$(f^m \smile g^n)(a_1, \cdots, a_m, b_1, \cdots, b_n) = f^m(a_1, \cdots, a_m) g^n(b_1, \cdots, b_n) .$$

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(3) Let A be a Hopf algebra, set $C^m = A \otimes_k A \otimes_k \cdots \otimes_k A$ (m times), and let $\Delta_{n-1}: A \to C^n$ be the map obtained by successive applications of the "diagonal map" Δ. Write $\Delta_{n-1}b = \sum_\alpha b_1^{(\alpha)} \otimes \cdots \otimes b_n^{(\alpha)}$, set

$$(a_1 \otimes \cdots \otimes a_i \otimes b \otimes a_{i+2} \otimes \cdots \otimes a_m) \circ_i (c_1 \otimes \cdots \otimes c_m)$$
$$= \sum_\alpha a_1 \otimes \cdots \otimes a_i \otimes b_1^{(\alpha)} c_1 \otimes \cdots \otimes b_n^{(\alpha)} c_n \otimes a_{i+2} \otimes \cdots \otimes a_m ,$$

and set

$$(a_1 \otimes \cdots \otimes a_m) \smile (c_1 \otimes \cdots \otimes c_n) = a_1 \otimes \cdots \otimes a_m \otimes c_1 \otimes \cdots \otimes c_n .$$

If $m = 0$ set

$$f^m \circ_i g^n = 0 .$$

A composition complex (c.c.) is *associative* if the cup product $\smile$ is associative. This is always so in *Examples* (1) and (3) and holds in (2) whenever the algebra A itself is associative. The complex is *unital* if there is a "unit element" $1 \in C^1$ such that $1 \circ_0 f^m = f \circ_i 1 = f$ for $0 \leqq i \leqq m-1$; henceforth this is tacitly assumed. In (2) the unit is the identity map of A, in (1) it is the cochain assigning to every ordered 1-simplex the value 1, and in (3) it is the unit element of A.

The elementary theory of composition complexes is essentially contained in [3], where *Example* (2) was considered in detail, but in which, in fact, only Axioms (a) and (b) are used. For the proofs of the assertions which follow, the reader may therefore refer to [3]. Set

$$f^m \circ g^n = \sum_{i=0}^{m-1} (-1)^{(n-1)i} f^m \circ_i g^n .$$

This "composition product" in (1) is identical, up to sign, with Steenrod's $\smile_1$; comparison with [13] shows that $f^m \smile_1 g^n = (-1)^{(m-1)n} f^m \circ g^n$. Setting

$$[f^m, g^n] = f^m \circ g^n - (-1)^{(m-1)(n-1)} g^n \circ f^m ,$$

one has

$$[f^m, g^n] = -(-1)^{(m-1)(n-1)} [g^n, f^m] ,$$
$$(-1)^{(m-1)(l-1)} [f^m, [g^n, h^l]] + (-1)^{(n-1)(m-1)} [g^n, [h^l, f^m]]$$
$$+ (-1)^{(l-1)(n-1)} [h^l, [f^m, g^n]] = 0 ,$$
$$[f^{2m+1}, f^{2m+1}] = 0 \quad \text{and} \quad [f^{2m}, [f^{2m}, f^{2m}]] = 0 .$$

Writing $(\operatorname{ad} \pi) f = [\pi, f]$, if $\pi \circ \pi = 0$, then $(\operatorname{ad} \pi)^2 = 0$. Define the coboundary operator $\delta: C^m \to C^{m+1}$ by $\delta f^m = (-1)^{m-1} [1 \smile 1, f]$. If $\mathcal{C}$ is associative, then

$$(1 \smile 1) \circ (1 \smile 1) = (1 \smile 1) \smile 1 - 1 \smile (1 \smile 1) = 0 ,$$

whence $\delta^2 = 0$. We may therefore define the groups of m-cocycles Z^m, m-coboundaries B^m, and the mth cohomology group $H^m = Z^m / B^m$ in the usual

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way, these coinciding, in *Examples* (1) and (2) with the usual ones. The cup product is defined on cohomology classes, and $H^m \smile H^n \subset H^{m+n}$.

In connection with the deformation theory of an associative algebra A, note in particular that we have the following formulas. If $f, g \in C^1$, then $(f \smile g)(a, b) = f(a)g(b)$, and $\delta f(a, b) = af(b) - f(ab) + f(a)b$. If $F, G \in C^2$, then

$$(F \circ G)(a, b, c) = F\big(G(a, b), c\big) - F\big(a, G(b, c)\big) ,$$

and

$$\delta F(a, b, c) = aF(b, c) - F(ab, c) + F(a, bc) - F(a, b)c .$$

Returning to the general case, even without the hypothesis of associativity, one has the basic identity

$$\begin{aligned} f^m \circ \delta g^n - \delta(f^m \circ g^n) + (-1)^{n-1}\delta f^m \circ g^n \\ = (-1)^{n-1}[g^n \smile f^m - (-1)^{mn} f^m \smile g^n] , \end{aligned} \tag{1.1}$$

from which it follows in the associative case that, if $\eta^m \in H^m$, $\zeta^n \in H^n$, then

$$\eta^m \smile \zeta^n = (-1)^{mn}\zeta^n \smile \eta^m . \tag{1.2}$$

From (1) it follows that, whether $\mathcal{C}$ is associative or not, one has $[Z^m, Z^n] \subset Z^{m+n-1}$, so there is a graded Lie structure on $Z^* = \bigoplus Z^m$. In the associative case one has $[Z^m, B^n] \subset B^{m+n-1}$, so there is a graded Lie structure on $H^* = \bigoplus H^m$, and one has, for all $\eta^m \in H^m$, $\zeta^n \in H^n$, and $\xi^l \in H^l$,

$$[\eta^m \smile \zeta^n, \xi^l] = [\eta^m, \xi^l] \smile \zeta^n + (-1)^{m(l-1)}\eta^m \smile [\zeta^n, \xi^l] . \tag{1.3}$$

Example 1 differs from the others in that, if $C^* = C^*(X, k)$, then the existence of the Steenrod $\smile_2$ shows that $[Z^m, Z^n] \subset B^{m+n-1}$, so the graded Lie structure on H^* is trivial. Note, in particular, that for every Hopf algebra A, we have defined cohomology groups $H^*(A)$ with both graded associative and graded Lie structures satisfying (2) and (3).

Since C^1 is always an associative algebra under the $\circ_0$ multiplication, we write fg, for simplicity, instead of $f \circ_0 g (= f \circ g)$, whenever $f \in C^1$.

The concept of a morphism of composition complexes is evident. A morphism, be it of rings or complexes, is always assumed to carry unit to unit.

A composition complex *truncated at dimension* N (c.c.N.), is a sequence $\mathcal{C} = \{C^0, C^1, \cdots, C^N\}$ of unital k-modules satisfying (a) and (b) to the extent to which they are meaningful. To the same extent, all the foregoing holds for a c.c.N. except that associativity is meaningless for a c.c.2.

2. Families of automorphisms and derivations

The basic concepts of the deformation theory introduced in [4] are meaningful for truncated composition complexes. An automorphism (more precise-

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ly, an "automorphism-like element") of a c.c.2 is an invertible element $\Phi \in C^1$ such that $\Phi(1 \smile 1) = \Phi \smile \Phi$, from which it follows that $\Phi(f \smile g) = \Phi f \smile \Phi g$ for all f and g. These form a group. A derivation is an element of Z^1, also denoted Der $\mathcal{C}$.

Let t be a variable, and let $\mathrm{Aut}_m \mathcal{C}$ denote the group of those elements of $C^1[t] = k[t] \otimes_k C^1$ of the form $\Phi_t = 1 + t\varphi_1 + t^2\varphi_2 + \cdots + t^m\varphi_m$, $\varphi_i \in C^1$, such that $\Phi_t(1 \smile 1) \equiv \Phi_t \smile \Phi_t \bmod t^{m+1}$. The necessary and sufficient condition for this is that, for all $l \leqq m$, we have (cf. [4])

$$\varphi_1 \smile \varphi_{l-1} + \varphi_2 \smile \varphi_{l-2} + \cdots + \varphi_{l-1} \smile \varphi_1 = -\delta\varphi_l \,. \tag{2.1}$$

Let $\mathrm{Aut}_\infty \mathcal{C}$ denote the group of formal power series $\Phi_t = 1 + t\varphi_1 + t^2\varphi_2 + \cdots$ with $\varphi_i \in C^1$ such that $\Phi_t(1 \smile 1) = \Phi_t \smile \Phi_t$; these are the "one-parameter families" of automorphisms of $\mathcal{C}$. There are canonical morphisms $\mathrm{Der}\,\mathcal{C} = \mathrm{Aut}_1 \mathcal{C} \leftarrow \mathrm{Aut}_2 \mathcal{C} \leftarrow \cdots \leftarrow \mathrm{Aut}_\infty \mathcal{C}$. An element $\Phi_t \in \mathrm{Aut}_m \mathcal{C}$ has a *prolongation* to $\mathrm{Aut}_n \mathcal{C}$ if there is an element in the latter group mapping onto Φ_t. Using the exponential mapping it is trivial that, if k contains the rationals $\mathbf{Q}$, then every $\varphi \in \mathrm{Der}\,\mathcal{C}$ can be prolonged to an element of $\mathrm{Aut}_\infty \mathcal{C}$, from which it follows that the same is true for any $\Phi_t \in \mathrm{Aut}_m \mathcal{C}$.

Let $\mathcal{C}$ now be an associative c.c.3. Then $\mathrm{Def}_m \mathcal{C}$ will denote the set of those elements of $C^2[t]$ of the form $\pi_t = 1 \smile 1 + tF_1 + t^2F_2 + \cdots + t^mF_m$, $F_i \in C^2$, such that $\pi_t \circ \pi_t \equiv 0 \bmod t^{m+1}$, while $\mathrm{Def}_\infty \mathcal{C}$ will denote the set of "one-parameter families of deformations", i.e., formal power series $\pi_t = 1 \smile 1 + tF_1 + t^2F_2 + \cdots$, such that $\pi_t \circ \pi_t = 0$. If $\mathcal{C} = C^*(A, A)$ where A is an associative k-algebra, then this is the necessary and sufficient condition that π_t define an associative multiplication on $A[[t]]$. There are canonical mappings $\{1 \smile 1\} = \mathrm{Def}_0 \mathcal{C} \leftarrow \mathrm{Def}_1 \mathcal{C} \leftarrow \cdots \leftarrow \mathrm{Def}_\infty \mathcal{C}$, and a concept of prolongation just as for automorphisms. The necessary and sufficient condition that π_t be in $\mathrm{Def}_m \mathcal{C}$ is that, for all $l \leqq m$, we have

$$F_1 \circ F_{l-1} + F_2 \circ F_{l-2} + \cdots + F_{l-1} \circ F_1 = \delta F_l \qquad \text{(cf. [4])} \,. \tag{2.2}$$

Setting $z(\pi_t) = F_1 \circ F_m + F_2 \circ F_{m-1} + \cdots + F_{m-1} \circ F_1$, $z(\pi_t)$ is a 3-cocycle whose cohomology class, denoted Obs π_t, vanishes if and only if π_t can be prolonged to an element of $\mathrm{Def}_{m+1} \mathcal{C}$. If $\mathcal{C} = C^*(X, k)$ and $\pi_t = 1 \smile 1 + tF$, then Obs π_t is the class of $F \circ F$, the Steenrod square of F.

Even though the general concepts of deformation and 2-cocycle are not defined for c.c.2's, we may make the following definitions. A *trivial deformation* of a c.c.2 is a series of the form, $\pi_t = \Phi_t^{-1}(\Phi_t \smile \Phi_t)$ where $\Phi_t = 1 + t\varphi_1 + t^2\varphi_2 + \cdots$, $\varphi_i \in C^1$. A series $\pi_t = 1 \smile 1 + tF_1 + t^2F_2 + \cdots$, $F_i \in C^2$, is a *restricted deformation* if $\pi_{tm} = 1 \smile 1 + t^mF_1 + t^{2m}F_2 + \cdots$ is trivial for some m, and F_1 is likewise called a *restricted 2-cocycle*. The k-submodule of

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C^2 spanned by these is denoted by RZ^2 and contains B^2. The *restricted second cohomology* group $RH^2 = RZ^2/B^2$ is thus well defined, in particular, for all algebras, and in the associative, Lie, and commutative associative cases, is a subgroup of the usual second cohomology group of the algebra with coefficients in itself as defined, respectively, in Hochschild [10], Chevalley-Eilenberg [2], and Harrison [9].

Let $\Phi_t = 1 + t\varphi_1 + \cdots + t^m\varphi_m$ be an element of $\mathrm{Aut}_m \mathcal{C}$. Setting $z(\Phi_t) = \varphi_1 \smile \varphi_m + \varphi_2 \smile \varphi_{m-1} + \cdots + \varphi_m \smile \varphi_1$, one has $z(\Phi_t) \in RZ^2$, and the cohomology class of $z(\Phi_t)$, denoted $\mathrm{Obs}\ \Phi_t$, vanishes if and only if Φ_t can be prolonged to an element of $\mathrm{Aut}_{m+1} \mathcal{C}$. Denoting by $\mathbf{Z}_p$ the localization at a prime p of the integers $\mathbf{Z}$, if there is a morphism $\mathbf{Z}_p \to k$, then any $\varphi \in \mathrm{Der}\, \mathcal{C}$ can be prolonged to $\Phi_t = 1 + t\varphi + (t^2/2!)\varphi^2 + \cdots + \big(t^{p-1}/(p-1)!\big)\varphi^{p-1}$, an element of $\mathrm{Aut}_{p-1} \mathcal{C}$. One then has, formally, $z(\Phi_t) = -(1/p!)\delta\varphi^p$. If k has characteristic p, then this is $(1/p)\delta\varphi^p$, which has been denoted by $\mathrm{Sq}_p \varphi$ in [4]. The foregoing is meaningful in particular when $\mathcal{C} = C^*\big(X, \mathbf{Z}/(p)\big)$; in which case it is easy to verify that, if X is a finite simplicial complex and $\varphi \in Z^1\big(X, \mathbf{Z}/(p)\big)$, then the cohomology class of $\mathrm{Sq}_p \varphi$ is the image of the class of φ under the Bockstein morphism $H^1\big(X, \mathbf{Z}/(p)\big) \to H^2\big(X, \mathbf{Z}/(p)\big)$ associated with the coefficient sequence

$$0 \longrightarrow \mathbf{Z}/(p) \longrightarrow \mathbf{Z}/(p^2) \longrightarrow \mathbf{Z}/(p) \longrightarrow 0\ .$$

3. The obstruction morphism theorem

THEOREM 1. *If $\mathcal{C}$ is a* c.c.2, *and if $\Phi_t, \Psi_t \in \mathrm{Aut}_m \mathcal{C}$, then* $\mathrm{Obs}\ \Phi_t\Psi_t = \mathrm{Obs}\ \Phi_t + \mathrm{Obs}\ \Psi_t$.

PROOF. Let

$$\Phi_t = 1 + t\varphi_1 + \cdots + t^m\varphi_m\ , \qquad \Psi_t = 1 + t\psi_1 + \cdots + t^m\psi_m\ ,$$

and set

$$c(\Phi_t, \Psi_t) = \varphi_1\psi_m + \varphi_2\psi_{m-1} + \cdots + \varphi_m\psi_1\ .$$

Omitting the parameter t for simplicity, it suffices to prove that $z(\Phi\Psi) - z(\Phi) - z(\Psi) = -\delta c(\Phi, \Psi)$. This is trivially true when Φ and Ψ have prolongations $\bar{\Phi} = 1 + t\varphi_1 + \cdots + t^{m+1}\varphi_{m+1}$ and $\bar{\Psi} = 1 + t\psi_1 + t^{m+1}\psi_{m+1}$, for then $z(\Phi) = -\delta\varphi_{m+1}$, $z(\psi) = -\delta\psi_{m+1}$, while the term of degree $m+1$ in $\bar{\Phi}\bar{\Psi}$, whose coboundary is $-z(\Phi\Psi)$, is $\varphi_{m+1} + \psi_{m+1} + c(\Phi, \Psi)$. Now let A be the ring $k\{x_1, \cdots, x_m, y_1, \cdots, y_m\}$ of polynomials in non-commuting variables $x_1, \cdots, y_m$, with coefficients in k. This can be made into a Hopf algebra in a unique way by setting $\Delta x_i = x_i \otimes 1 + x_{i-1} \otimes x_1 + x_{i-2} \otimes x_2 + \cdots + 1 \otimes x_i$, and similarly for Δy_i, and the tensor algebra over A is then a c.c. as shown in § 1, *Example* 3. Let $\mathcal{B}(m)$ denote the c.c.2. obtained by truncating this at dimension 2, and set

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$$\Phi' = 1 + tx_1 + \cdots + t^m x_m , \qquad \Psi' = 1 + ty_1 + \cdots + t^m y_m .$$

These are in $\mathrm{Aut}_m \mathcal{B}(m)$. There is a canonical monomorphism $\mathcal{B}(m) \to \mathcal{B}(m+1)$ the images under which of Φ' and Ψ' can be prolonged, whence $z(\Phi'\Psi') - z(\Phi') - z(\Psi') = -\delta c(\Phi', \Psi')$. On the other hand, there is a morphism $\mathcal{B}(m) \to \mathcal{C}$ defined by $x_i \mapsto \varphi_i$, $y_i \mapsto \psi_i$, $i = 1, \cdots, m$, and this morphism carries $z(\Phi')$ to $z(\Phi)$, and similarly for $z(\Psi')$, $z(\Phi'\Psi')$ and $c(\Phi', \Psi')$. This ends the proof.

A generalization of this theorem is valid for filtered rings [8].

Henceforth, if there is a morphism $\mathbf{Q} \to k$ or $\mathbf{Z}_p \to k$ then a c.c. with coefficients in k will be called a c.c. "over $\mathbf{Q}$" or a c.c. "over $\mathbf{Z}_p$" respectively, k being understood.

4. The obstruction exponent

Let $\mathcal{C}$ be a c.c.2. and $\Phi = 1 + t\varphi_1 + \cdots + t^m\varphi_m$ be an elemeut of $\mathrm{Aut}_m \mathcal{C}$. If there is an integer n, such that Φ can be prolonged in some way to an element of $\mathrm{Aut}_{n-1} \mathcal{C}$ but such that no prolongation to $\mathrm{Aut}_n \mathcal{C}$ exists, then Φ will be said to have *obstruction order* n. In the contrary case, Φ will be said to have obstruction order ∞, but this does not necessarily imply that Φ can be prolonged to an element of $\mathrm{Aut}_\infty \mathcal{C}$, since every prolongation of Φ may ultimately be obstructed. If $\Phi = 1 + t^m\varphi$, then (2.1) implies that $\delta\varphi = 0$, and conversely if $\delta\varphi = 0$, then $1 + t^m\varphi \in \mathrm{Aut}_m \mathcal{C}$.

THEOREM 2. *Let $\mathcal{C}$ be a* c.c.2. *over $\mathbf{Z}_p$ and φ be an element of Z^1. If the obstruction order of $1 + t^m\varphi$ is finite, then it is of the form $n = mp^e$ for some integer $e = e(\varphi, m) > 0$.*

PROOF. Fix an integer $n > m$, let G denote the subgroup of $\mathrm{Aut}_{n-1} \mathcal{C}$ consisting of all elements of the form $\Phi = 1 + t^m\varphi_m + \cdots + t^{n-1}\varphi_{n-1}$, and G' be the normal subgroup consisting of all elements of the form $1 + t^{m+1}\psi_{m+1} + \cdots + t^{n-1}\psi_{n-1}$. The images of G and G' under the morphism Obs: $\mathrm{Aut}_{n-1}\mathcal{C} \to H^2$ will be denoted $\mathrm{Obs}\, G$ and $\mathrm{Obs}\, G'$, respectively. The assertion of the theorem is that $\mathrm{Obs}\, G/\mathrm{Obs}\, G'$ reduces to the zero element unless $n = mp^e$ for some e. The image of $\mathrm{Obs}\, \Phi$ in $\mathrm{Obs}\, G/\mathrm{Obs}\, G'$ depends only on φ_m and will be denoted $\mathrm{obs}\, \varphi_m$. One may view obs as a morphism from a subgroup of Z^1 to a quotient group of H^2, namely $H^2/\mathrm{Obs}\, G'$.

Let k^m now denote the subring of k generated by all elements of the form α^m, $\alpha \in k$, and similarly for k^n. By Theorem 1, Obs is additive, and has the further obvious property that $\mathrm{Obs}\, \Phi_{\alpha t} = \alpha^n \mathrm{Obs}\, \Phi_t$ for all $\alpha \in k$. This implies that

$$\mathrm{obs}\, \alpha^m\varphi_m = \alpha^n \mathrm{obs}\, \varphi_m \tag{4.1}$$

for all $\alpha \in k$. Set $\varphi_m = \varphi$, and let I be the annihilator of obs φ in k. It fol-

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lows from (4.1) that there is a ring epimorphism $k^m \longrightarrow k^n/I$ defined by $\alpha^m \mapsto \alpha^n \bmod I$. The assertion of the theorem is that this is the zero morphism unless $n = mp^e$ for some e.

We show first that it is sufficient to consider the case where k is of characteristic p.

For suppose the theorem true in this case. Then the ideal (I, p) generated by I and p in k^n is all of k^n, and therefore p is invertible in k^n/I. Thus, if $k^n/I \neq 0$, then $k^n/I \supset \mathbf{Q}$, whence p is neither the zero element of k nor a nilpotent element. It follows that the prime ring of k contains $\mathbf{Z}$, and that the restriction to $\mathbf{Z}$ of the morphism $k^m \longrightarrow k^n/I$ is an isomorphism. This is impossible.

The proof is therefore reduced to the case where k is of characteristic p. If k contains a field k_0 algebraic over $\mathbf{Z}/(p) = \mathbf{F}_p$, then k_0^m and k_0^n are both fields, and the mapping $\alpha^m \mapsto \alpha^n$ can be extended to an isomorphism of k_0^m with k_0^n. Suppose $n \neq mp^e$ for any e. Then this puts an absolute bound on the number of elements in k_0. Therefore, setting $q = p^d$ with large d, and letting the extension of $\varphi = \varphi_m$ to $\mathcal{C}^* = \mathcal{C} \otimes_{\mathbf{F}_p} \mathbf{F}_q$ be denoted still by φ_m, we have $\operatorname{obs} \varphi_m = 0$ relative to $\mathcal{C}^*$. That is, there exists a $\Phi^* = 1 + t^m\varphi_m + t^{m+1}\varphi_{m+1}^* + \cdots + t^n\varphi_n^*$ in $\operatorname{Aut}_n \mathcal{C}^*$, where $\varphi_i^* \in C^1 \otimes \mathbf{F}_q$. The proof will be at an end if we can show that a Φ^* can be found in which the φ_i^* are all defined over k. Now we may assume that the large integer d has been chosen prime to p. Let σ be a generator of the Galois group of $\mathbf{F}_q$ over $\mathbf{F}_p$ and set

$$\Psi^* = \prod_{i=1}^{d} \sigma^i \Phi^* ,$$

the order of the product being unimportant. If j is the largest integer such that φ_j^* is defined over k, and if we write $\Psi^* = 1 + t^m\psi_m^* + \cdots + t^n\psi_n^*$, then ψ_i^* is defined over k for $m \leqq i \leqq j + 1$. Since $p \nmid d$, and k is a $\mathbf{Z}_p$-module, $(\Psi^*)^{1/d}$ is well-defined. But $\psi_m^* = d\varphi_m$, so $(\Psi^*)^{1/d}$ begins $1 + t^m\varphi_m + \cdots$. It follows by induction that we can construct an element of $\operatorname{Aut}_n \mathcal{C}^*$ beginning $1 + t^m\varphi_m + \cdots$, all of whose coefficients are defined over k. This ends the proof.

It will be shown that the integer $e(\varphi, m)$ of Theorem 2 is actually independent of m. Observe first that $e(\varphi, mm') \geqq e(\varphi, m)$, for if $1 + t^m\varphi$ can be prolonged to an element of $\operatorname{Aut}_{n-1} \mathcal{C}$, then replacing t by $t^{m'}$ shows that $1 + t^{mm'}\varphi$ can be prolonged to an element of $\operatorname{Aut}_{nm'-1}\mathcal{C}$. To prove the independence of $e(\varphi, m)$ of m, it is therefore sufficient to prove that $e(\varphi, 1) \geqq e(\varphi, m)$ for all m. The essential step is the following.

THEOREM 3. *Let $\mathcal{C}$ be a unital* c.c.2. *over $\mathbf{Z}_p$ and Φ be an element of* $\operatorname{Aut}_n \mathcal{C}$ *of the form*

$$\Psi = 1 + t^m\varphi_m + t^{m+1}\varphi_{m+1} + \cdots + t^n\varphi_n .$$

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Set $l = [n/m]$, the greatest integer contained in n/m, and let j be the least integer, such that both $m \nmid j$ and $\varphi_j \neq 0$. Then

(1) $\varphi_j \in Z^1$ *and*

(2) Φ *has a factorization* $\Phi = \Phi'\Phi''$, *where* $\Phi', \Phi'' \in \mathrm{Aut}_n \mathcal{C}$, *and* Φ', Φ'' *have the following forms*

$$\Phi' = 1 + t^m\varphi_m + t^{2m}\varphi'_{2m} + \cdots + t^{lm}\varphi'_{lm} ,$$

a polynomial in t^m, and

$$\Phi'' = 1 + t^j\varphi_j + t^{j+1}\varphi''_{j+1} + \cdots + t^n\varphi''_n .$$

PROOF. In (2.1) with $l = j$ every product on the left has one factor zero, so $\delta\varphi_j = 0$. Suppose now that the theorem were false. Consider the least n for which it fails, for that n, consider the largest m, and for that n and m, consider the largest j. Then

$$\Phi_{n-1} = 1 + t^m\varphi_m + \cdots + t^{n-1}\varphi_{n-1}$$

is an element of $\mathrm{Aut}_{n-1} \mathcal{C}$, and by the inductive hypothesis can be represented in the form $\Phi_{n-1} = \Phi'_{n-1}\Phi''_{n-1}$, where Φ'_{n-1} is a polynormial in t^m and $\Phi''_{n-1} = 1 + t^j\varphi_j + t^{j+1}\varphi''_{j+1} + \cdots + t^{n-1}\varphi''_{n-1}$. Now either n/j or n/m is not a power of p. In the former case, by Theorem 2, there exists an element of $\mathrm{Aut}_n \mathcal{C}$ of the form

$$\Psi = 1 + t^j\varphi_j + t^{j+1}\psi_{j+1} + \cdots + t^n\psi_n .$$

Then $\Phi\Psi^{-1}$ is an element of $\mathrm{Aut}_n \mathcal{C}$ for which the value of j has been increased by one. Hence it can be factored, from which one gets the required factorization of Φ. In the latter case, consider t^m as the variable in Φ'_{n-1}. It follows again from Theorem 2 that there exists an element of $\mathrm{Aut}_n \mathcal{C}$ of the form $\Psi = 1 + t^m\varphi_m + t^{2m}\psi_{2m} + \cdots$. In this case, $\Psi^{-1}\Phi$ is an element of $\mathrm{Aut}_n\mathcal{C}$ for which the value of m has been increased. If $2m > n$, then we can set $\Phi' = \Psi$, $\Phi'' = \Psi^{-1}\Phi$, otherwise, factoring $\Psi^{-1}\Phi$ gives the desired result and ends the proof.

In the Φ' of the preceding theorem, one can replace t^m by t. Therefore, if one has an element $\Phi = 1 + t^m\varphi + t^{m+1}\varphi_{m+1} + \cdots + t^n\varphi_n$ in $\mathrm{Aut}_n \mathcal{C}$, and if $n \geqq lm$ for some integer l, then there exists an element of $\mathrm{Aut}_l \mathcal{C}$ of the form $1 + t\varphi + t^2\psi_2 + \cdots + t^l\psi_l$. If $n = mp^e$, then taking $l = p^e$ shows that $e(\varphi, 1) \geqq e(\varphi, m)$. This proves

COROLLARY 1. *If $\varphi \in Z^1$, then $e(\varphi, m)$ is independent of m.*

We write $e(\varphi)$ for this integer and call it the *obstruction exponent* of φ; it may be $+\infty$. Since the theorem holds in particular for complexes constructed as in the proof of Theorem 1, we have

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COROLLARY 2. *Let* $q = p^n$ *and* $\Phi \in \mathrm{Aut}_{mq}\, \mathcal{C}$ *be of the form* $1 + t^m\varphi_m + \cdots + t^{mq}\varphi_{mq}$. *Then there exist polynomials* $y_2(x), \cdots, y_{pq-1}(x) \in k\{x_m, x_{m+1}, \cdots, x_{mq}\}$ *such that* $1 + t\varphi_m + t^2 y_2(\varphi) + \cdots + t^{pq-1}y_{pq-1}(\varphi) \in \mathrm{Aut}_{pq-1}\, \mathcal{C}$.

It is an open question if an element $1 + t^m\varphi_m + t^{m+1}\varphi_{m+1} + \cdots$ in $\mathrm{Aut}_\infty \mathcal{C}$ has a factorization of the form given in Theorem 3.

5. The filtration of Der $\mathcal{C}$

Let $\mathcal{C}$ be a c.c.2. over $\mathbf{Z}_p$, and set $\mathrm{Der}_{n-1}\mathcal{C} = \{\varphi \in \mathrm{Der}\,\mathcal{C} \mid e(\varphi) \geqq n\}$, $\mathrm{Der}_\infty \mathcal{C} = \bigcap \mathrm{Der}_n\, \mathcal{C}$. Those $\varphi \in \mathrm{Der}_\infty\, \mathcal{C}$ for which there exists an element of $\mathrm{Aut}_\infty\, \mathcal{C}$ beginning $1 + t^m\varphi + \cdots$ form a subgroup denoted $\mathrm{Der}_\omega\, \mathcal{C}$, but it is not known if it is a k-submodule of $\mathrm{Der}_\infty\, \mathcal{C}$.

THEOREM 4. *If* $\mathcal{C}$ *is a* c.c.2. *over* $\mathbf{Z}_p$, *then* $\mathrm{Der}_n\, \mathcal{C}$ *is a Lie subalgebra of* $\mathrm{Der}\,\mathcal{C}$ *for all finite* n *and for* $n = \infty$, *and* $\mathrm{Der}_\omega\, \mathcal{C}$ *is a Lie subring of* $\mathrm{Der}\,\mathcal{C}$. *If* $\mathcal{C}$ *has characteristic* p, *then* $\mathrm{Der}_n\, \mathcal{C}$ *is restricted for all* n *including* $n = \infty$ *and* $n = \omega$.

PROOF. The assertions for $\mathrm{Der}_\omega\, \mathcal{C}$ are trivial and those for $\mathrm{Der}_\infty\, \mathcal{C}$ follow from those for $\mathrm{Der}_n\, \mathcal{C}$ with n finite, so we may suppose that n is finite, and set $q = p^n$. If $\varphi, \psi \in \mathrm{Der}_{n-1}\, \mathcal{C}$, then by hypothesis, there exist elements $\Phi = 1 + t\varphi + \cdots + t^{q-1}\varphi_{q-1}$ and $\Psi = 1 + t\psi + \cdots + t^{q-1}\psi_{q-1}$ in $\mathrm{Aut}_{q-1}\, \mathcal{C}$. It follows from Theorem 1 that $\mathrm{Obs}\, \Phi\psi\Phi^{-1}\psi^{-1} = 0$, whence there exists an element of $\mathrm{Aut}_q\, \mathcal{C}$ of the form $1 + t^2[\varphi, \psi] + t^3\theta_3 + \cdots + t^q\theta_q$. It follows from the results of the previous section that $[\varphi, \psi] \in \mathrm{Der}_{n-1}\mathcal{C}$. If the characteristic is p, then $\Phi^p = 1 + t^p\varphi^p + \cdots$. Since $\mathrm{Obs}\, \Phi^p = p\, \mathrm{Obs}\, \Phi = 0$, it follows that $\varphi^p \in \mathrm{Der}_{n-1}\, \mathcal{C}$

THEOREM 5. *Let* $\mathcal{C}$ *be a unital* c.c.2. *over* $\mathbf{Z}_p$. *If for some finite* e *it is the case that* $\mathrm{Der}_e\, \mathcal{C} = \mathrm{Der}_\infty\, \mathcal{C}$, *then* $\mathrm{Der}_e\, \mathcal{C} = \mathrm{Der}_\infty\, \mathcal{C} = \mathrm{Der}_\omega\, \mathcal{C}$.

PROOF. Set $p^e = q$. It is sufficient to show for every $m > 1$ that, if $\Phi = 1 + t\varphi_1 + \cdots + t^{mq}\varphi_{mq}$ is an element of $\mathrm{Aut}_{mq}\, \mathcal{C}$, then for every $N > m$, $1 + t\varphi_1 + \cdots + t^m\varphi_m$ can be prolonged to an element of $\mathrm{Aut}_N\, \mathcal{C}$. This is true, by hypothesis, for $m = 1$. Suppose it true for some $m - 1$. Then given Φ and N, we can prolong $1 + t\varphi_1 + \cdots + t^{m-1}\varphi_{m-1}$ to an element of $\mathrm{Aut}_N\, \mathcal{C}$ of the form $\Psi = 1 + t\varphi_1 + \cdots + t^{m-1}\varphi_{m-1} + t^m\psi_m + \cdots + t^N\psi_N$. Now $\Psi^{-1}\Phi \bmod t^{mq+1}$ is a polynomial of the form $1 + t^m(\varphi_m - \psi_m) + t^{m+1}\theta_{m+1} + \cdots + t^{mq}\theta_{mq}$, so $\varphi_m - \psi_m \in \mathrm{Der}_e\, \mathcal{C}$, whence by hypothesis, there exists an element of $\mathrm{Aut}_N\, \mathcal{C}$ of the form

$$P = 1 + t^m(\varphi_m - \psi_m) + t^{m+1}\rho_{m+1} + \cdots + t^N\rho_N\,.$$

Thus ΨP is an element of $\mathrm{Aut}_N\, \mathcal{C}$ of the form $1 + t\varphi_1 + \cdots + t^m\varphi_m + \cdots$, and the induction proceeds, ending the proof.

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The hypothesis of the theorem holds if, in addition to being a $\mathbf{Z}_p$-algebra, k is noetherian and $\mathrm{Der}\,\mathcal{C}/\mathrm{Der}_\omega\,\mathcal{C}$ is of finite type over k. If A is an associative algebra with unit, and $\mathcal{C} = C^*(A, A)$, then $B^1(A, A) \subset \mathrm{Der}_\omega A$, so it is sufficient that k be noetherian and $H^1(A, A)$ be of finite type.

6. The filtration of RH^2 and the group $\mathrm{Aut}_*\,\mathcal{C}$

Let $\mathcal{C}$ be a c.c.2. over $\mathbf{Z}_q$. Given an integer $e \geqq 0$, set $q = p^{e+1}$ and let R_eZ^2 denote the k-module generated by those $F \in Z^2$ for which there exists a trivial one-parameter family of deformations of $\mathcal{C}$ of the form $1 \frown 1 + t^qF + t^{q+1}F_{q+1} + \cdots$. Define $R_{-1}Z^2 = \delta C^1 = B^2$. Replacing t by t^p shows that $R_eZ^2 \subset R_{e+1}Z^2$ for all e.

THEOREM 6. *Let $\mathcal{C}$ be a* c.c.2. *over $\mathbf{Z}_p$, and $\Phi_t = 1 + t^j\varphi_j + \cdots + t^{n-1}\varphi_{n-1}$ be an element of $\mathrm{Aut}_{n-1}\,\mathcal{C}$ with $n/j < p^{e+1}$ for some e. Then $z(\Phi_t) \in R_{e-1}Z^2$.*

PROOF. For $e = 0$, the assertion is that Φ_t can be prolonged to an element of $\mathrm{Aut}_n\,\mathcal{C}$, which is evident. Suppose now that the theorem is false. Choose the least e for which it fails, for that e choose the least n, and for that e and n choose the largest j. We may without loss of generality assume that $p^e \leqq n/j$. The existence of Φ_t implies that $e(\varphi_j) \geqq e$, whence there exists an element of $\mathrm{Aut}_{q-1}\,\mathcal{C}$ of the form $\Psi_t = 1 + t\varphi_j + t^2\psi_2 + \cdots + t^{q-1}\psi_{q-1}$. Moreover, $\Psi_{tj} = 1 + t^j\varphi_j + t^{2j}\psi_2 + \cdots + t^{j(q-1)}\psi_{q-1}$ is an element of $\mathrm{Aut}_{jq-1}\mathcal{C}$. Set $\Phi_t = \Psi_{tj} \bmod t^n$. Then we can write $\Phi_t = \Phi_t'\Phi_t''$ with Φ_t'' of the form $\Phi_t'' = 1 + t^{j+1}\varphi_{j+1}'' + \cdots$. Since $z(\Phi_t) = z(\Phi_t') + z(\Phi_t'')$ + coboundary, and since the inductive hypothesis applies to $z(\Phi_t'')$ and also to the coboundary, in order to prove the theorem, it remains only to show that $z(\Phi_t') \in R_{e-1}Z^2$. If $n < jq - 1$, then $z(\Phi_t') = 0$, while if $n = jq - 1$, then $z(\Phi_t') = z(\Psi_t)$. Since $\Psi_t^{-1}(\Psi_t \frown \Psi_t)$ is a trivial one-parameter family of deformations of $\mathcal{C}$ beginning $1 \frown 1 + t^qz(\Psi_t) + \cdots$, this shows that $z(\Phi_t') \in R_{e-1}Z^2$, ending the proof.

Since every restricted 2-cocycle is of the form $z(\Phi)$ for some Φ, it follows that, if $\mathcal{C}$ is a unital c.c.2. over $\mathbf{Z}_p$, then $RZ^2 = \bigcup R_eZ^2$, the union being extended over all positive e. There is thus a natural filtration of RH^2 defined by setting $R_eH^2 = R^eZ^2/B^2$.

For every m and n, there exists a canonical monomorphism $\mathrm{Aut}_{m-1}\,\mathcal{C} \longrightarrow \mathrm{Aut}_{mn-1}\,\mathcal{C}$ defined by

$$1 + t\varphi_1 + \cdots + t^{m-1}\varphi_{m-1} \longrightarrow 1 + t^n\varphi_1 + t^{2n}\varphi_2 + \cdots + t^{(m-1)n}\varphi_{m-1}\,,$$

the latter being considered as a polynomial in t of degree $mn - 1$. Denote the direct limit of the groups $\mathrm{Aut}_m\,\mathcal{C}$ by $\lim \mathrm{dir}\,\mathrm{Aut}\,\mathcal{C}$, and the image of $\mathrm{Aut}_m\,\mathcal{C}$ in the direct limit by $\lim \mathrm{dir}\,\mathrm{Aut}_m\,\mathcal{C}$. The subgroup P of $\mathrm{Aut}\,\mathcal{C}$ consisting of the images of all elements in all $\mathrm{Aut}_m\,\mathcal{C}$ which have a prolongation to $\mathrm{Aut}_{m+1}\mathcal{C}$

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is normal by Theorem 1. Denote the quotient $(\lim \mathrm{dir}\, \mathrm{Aut}\mathcal{C})/P$ by $\mathrm{Aut}_*\,\mathcal{C}$ and the canonical morphism $\mathrm{Aut}_m\,\mathcal{C} \to \mathrm{Aut}_*\,\mathcal{C}$ by i_*. By the results of § 4, if $\mathcal{C}$ is a c.c.2. over $\mathbf{Z}_p$, then

$$\mathrm{Aut}_*\,\mathcal{C} = \bigcup_e i_*\,\mathrm{Aut}_{p^e-1}\,\mathcal{C}\ .$$

The group $\mathrm{Aut}_*\,\mathcal{C}$ consists of classes of obstructed "approximate automorphisms", and is abelian, for if $\sigma, \tau \in \mathrm{Aut}_m\mathcal{C}$, then $\mathrm{Obs}\,(\sigma\tau\sigma^{-1}\tau^{-1}) = 0$. Set $\mathrm{gr}_e\,\mathrm{Aut}_*\,\mathcal{C} = i_*\,\mathrm{Aut}_{p^{e+1}-1}\,\mathcal{C}/i_*\,\mathrm{Aut}_{p^e-1}\,\mathcal{C}$, $e = 0, 1, \cdots$, and set

$$\mathrm{gr}\,\mathrm{Aut}_*\,\mathcal{C} = \bigoplus_{e=0}^{\infty} \mathrm{gr}_e\,\mathrm{Aut}_*\,\mathcal{C}\ .$$

Every element of $\mathrm{gr}_e\,\mathrm{Aut}_*\,\mathcal{C}$ is represented by an element $\Phi = 1 + t\varphi + \cdots + t^{q-1}\varphi_{q-1}$ of $\mathrm{Aut}_{q-1}\mathcal{C}$, where $q = p^{e+1}$. The map $\Phi \mapsto \varphi$ is an isomorphism of $\mathrm{gr}_e\,\mathrm{Aut}_*\,\mathcal{C}$ with $\mathrm{gr}_e\,\mathrm{Der}\,\mathcal{C} = \mathrm{Der}_e\,\mathcal{C}/\mathrm{Der}_{e+1}\mathcal{C}$. Setting $\mathrm{gr}_e\,RH^2 = R_eH^2/R_{e-1}H^2$, there exist canonical isomorphisms

$$\mathrm{gr}_e\,\mathrm{Der}\,\mathcal{C} \simeq \mathrm{gr}_e\,\mathrm{Aut}_*\,\mathcal{C} \simeq \mathrm{gr}_e\,RH^2\ .$$

The canonical group isomorphism $\mathrm{gr}_e\,\mathrm{Der}\,\mathcal{C} \to \mathrm{gr}_e\,RH^2$ will be denoted by obs_e. One has $\mathrm{obs}_e\,\lambda\varphi = \lambda^{p^{e+1}}\,\mathrm{obs}_e\,\varphi$ for all $\lambda \in k$. We may therefore define a new k-module structure on $\mathrm{gr}_e\,RH^2$ by setting $\lambda\cdot\zeta = \lambda^{p^{e+1}}\zeta$ for all $\zeta \in \mathrm{gr}_e\,RH^2$. With this definition, obs_e becomes a k-module morphism, and there is thus a k-module isomorphism

$$\mathrm{gr}\,\mathrm{Der}\,\mathcal{C} \xrightarrow{\sim} \mathrm{gr}\,RH^2\ , \tag{6.1}$$

where on the right the k-module structure has been redefined separately on each direct summand. When k is a perfect field of characteristic p, it follows that

$$\dim_k\,(\mathrm{Der}\,\mathcal{C}/\mathrm{Der}_\infty\,\mathcal{C}) = \dim_k RH^2\ .$$

If $\mathrm{Der}_0\,\mathcal{C} = \mathrm{Der}_1\,\mathcal{C} = \cdots = \mathrm{Der}_e\,\mathcal{C}$, then $R_iH^2 = 0$ for $i = 0, 1, \cdots, e-1$. In this case there exists a canonical isomorphism $\mathrm{Der}_e\,\mathcal{C}/\mathrm{Der}_{e+1}\,\mathcal{C} \simeq R_eH^2$, so every element $\varphi \in \mathrm{Der}_e\,\mathcal{C}$ has a well-defined obstruction, namely its image denoted $\mathrm{Obs}\,\varphi$ under the composite morphism $\mathrm{Der}_e\,\mathcal{C} \to \mathrm{Der}_e\,\mathcal{C}/\mathrm{Der}_{e+1}\,\mathcal{C} \to R_eH^2$; $\mathrm{Obs}\,\varphi$ vanishes if and only if $\varphi \in \mathrm{Der}_{e+1}\,\mathcal{C}$.

THEOREM 7. *Let $\mathcal{C}$ be a* c.c.2. *over $\mathbf{Z}_p$ and suppose that $\mathrm{Der}_0\,\mathcal{C} = \mathrm{Der}_e\,\mathcal{C}$. Let φ be an element of $\mathrm{Der}_e\,\mathcal{C}$ such that $\mathrm{Obs}\,\varphi \neq 0$, and let F be an element of RZ^2 representing $\mathrm{Obs}\,\varphi$. Then there exists a restricted one-parameter family of deformations of $\mathcal{C}$ of the form $\pi_t = 1 \frown 1 + tF + t^2F_2 + \cdots$. Further π_{t^q} is trivial for $q = p^{e+1}$, but not for any smaller q.*

PROOF. It is sufficient to show that there exists a trivial deformation of the form $1 \frown 1 + t^qF + t^{2q}F_2 + \cdots$. For were the deformation still trivial when t^q was replaced by t^n for any $n < q$, then we would have $F \in R_{e-1}Z^2$,

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which is impossible. Now by hypothesis there exists a trivial deformation of the form $\Phi_t^{-1}(\Phi_t \smile \Phi_t) = f_t = 1 \smile 1 + t^q F + t^{q+1} G_{q+1} + \cdots$. Let j be the least integer such that $q \nmid j$ and $G_j \neq 0$. Set $k' = k[x]/(x^q - 1)$, and denote by ζ the image of x in k'. Then $\zeta^q = 1$. Setting $\Psi_t = \Phi_t \Phi_{\zeta t}^{-1}$, one has

$$\Psi_t^{-1}(\Psi_t \smile \Psi_t) = 1 \smile 1 + t^j(1 - \zeta^j)G_j + \cdots .$$

But $q \nmid j$ and $\operatorname{Der} \mathcal{C} = \operatorname{Der}_e \mathcal{C}$. Therefore, $(1 - \zeta^j)G_j \in B^2 \otimes_k k'$. Since $k' = k[\zeta]$ is a free k-module, it follows that $G_j \in B^2$. Let $G_j = \delta\gamma_j$ for some $\gamma_j \in C^1$. Set $\Phi_t' = \Phi_t(1 + t^j\gamma_j)$, and set $f_t' = \Phi_t'^{-1}(\Phi_t' \smile \Phi_t')$. Then the coefficients of $t, t^2, \cdots, t^{j-1}$ in f_t' all coincide with those in f_t, but the coefficient of t^j in f_t' is zero. Continuing in this way, we may construct a trivial formal one-parameter family of deformations of $\mathcal{C}$ of the form $1 \smile 1 + t^q F + t^{2q} F_2 + \cdots$, as desired.

COROLLARY. *Either* $\operatorname{Der} \mathcal{C} = \operatorname{Der}_\omega \mathcal{C}$ *or there exists a restricted one-parameter family of deformations of* $\mathcal{C}$ *which is not trivial.*

CHAPTER II. EXPLICIT DEFORMATION FORMULAS

1. A deformation formula for composition complexes over Q

If $\mathcal{C}$ is a c.c.2. and S a commutative subalgebra of C^1, then one can define a multiplication in $S \cup S$, the submodule of C^2 spanned by all $f \cup g$ with $f, g \in S$, by setting $(f \cup g)(f' \cup g') = ff' \cup gg'$. In general, this multiplication can not be extended to all of $C^1 \cup C^1$. The following lemma is basic.

LEMMA 1. *Let* $\mathcal{C}$ *be an associative* c.c.3. *over* $\mathbf{Q}$ *and* φ, ψ *be commuting elements of* $\operatorname{Der} \mathcal{C}$. *Set* $\pi_t = \exp t\varphi \smile \psi = 1 \smile 1 + t\varphi \smile \psi + (t^2/2!)\varphi^2 \smile \psi^2 + \cdots$. *Then* $\pi_t \circ \pi_t = 0$, i.e., π_t *is a one-parameter family of deformations of* $\mathcal{C}$.

PROOF. Compute.

Keeping the hypotheses, let $\mathcal{C}_t$ be the c.c.3, the ith module of which is the $k[[t]]$-module $C^i[[t]]$, composition being that induced from $\mathcal{C}$ but with cup product defined by π_t. One may verify readily that φ, ψ continue to be derivations of $\mathcal{C}_t$, and the same is true of any derivations commuting with both φ and ψ. Thus if $\varphi, \psi, \varphi', \psi'$ are commuting derivations of $\mathcal{C}$, and if t, t' are variables, then $\exp t'\varphi' \smile \psi'$ is a one-parameter family of deformations of $\mathcal{C}_t$. The $k[[t, t']]$ modules $C^i[[t, t']]$ become a c.c.3. with cup product defined by $\pi_{t,t'} = \exp(t\varphi \smile \psi + t'\varphi' \smile \psi')$, the composition structure being unchanged. One can specialize t' to be some power, t^n, of t. Repeated application yields

THEOREM 8. *Let* $\mathcal{C}$ *be a* c.c.2. *over* $\mathbf{Q}$, φ_i, ψ_i, $i = 1, 2, \cdots$ *be commuting derivations of* $\mathcal{C}$, *and* n_i *be a sequence of positive integers tending to* $+\infty$. *Then* $\pi_t = \exp \sum t^{n_i} \varphi_i \smile \psi_i$ *is a one-parameter family of deformations of* $\mathcal{C}$.

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2. Formal exponentials

In a composition complex over $\mathbf{Q}$, it is readily verified that, if $\theta \in C^1$, then $e^{-t\theta}(e^{t\theta} \smile e^{t\theta}) = e^{t \operatorname{ad} \theta}(1 \smile 1)$. If $\Phi_t = 1 + t\varphi_1 + t^2\varphi_2 + \cdots + t^n\varphi_n$ is an element of $\operatorname{Aut}_n \mathcal{C}$, set $\Phi_t = 1 + x$ with $x = t\varphi_1 + t^2\varphi_2 + \cdots + t^n\varphi_n$, and formally, $\Phi_t = \exp \Theta_t$, where $\Theta_t = x - x^2/2 + x^3/3 - \cdots$. Then $\Theta_t = t\theta_1 + t^2\theta_2 + \cdots$, where $\theta_1 = \varphi_1$ and θ_i is a polynomial in $\varphi_1, \cdots, \varphi_i$ for $i = 1, \cdots, n$, and a polynomial in $\varphi_1, \cdots, \varphi_n$ for $i > n$. The coefficients of the θ_i are not integers, so these formal polynomials are regarded as elements of the free non-commutative polynomial ring over $\mathbf{Q}$ with generators $[x], [y], \cdots$ corresponding to the non-zero elements $x, y, \cdots$ of C^1, reduced modulo the relations $[xy] - [x][y] = 0$, and $[0] = 0$. No confusion will arise if we write simply x for $[x]$ when such a formal polynomial is understood. In the foregoing, if the φ_i commute, then Φ_t will be called *abelian*. The θ_i then commute, and we have

$$\begin{aligned}\Phi_t^{-1}(\Phi_t \smile \Phi_t) &= (\exp \operatorname{ad} \Theta_t)(1 \smile 1) \\ &= [\exp (t \operatorname{ad} \theta_1 + t^2 \operatorname{ad} \theta_2 + \cdots)](1 \smile 1) .\end{aligned}$$

If $\delta\theta_1 = \delta\theta_2 = \cdots = \delta\theta_m = 0$ with $m < n$, then $[\exp (t \operatorname{ad} \theta_i)](1 \smile 1) = 1 \smile 1$ for $i \leqq m$, so $\Phi_t^{-1}(\Phi_t \smile \Phi_t) = 1 \smile 1 + t^{m+1}\delta\theta_{m+1} + \cdots$, whence $\delta\theta_{m+1} = 0$. Therefore $\delta\theta_1 = \delta\theta_2 = \cdots = \delta\theta_n = 0$, and the expansion of $\Phi_t^{-1}(\Phi_t \smile \Phi_t)$ begins with $1 \smile 1 + t^{n+1}\delta\theta_{n+1} + \cdots$. Since $\Phi_t^{-1}(\Phi_t \smile \Phi_t)$ has coefficients which are integral polynomials in the φ_i, it follows that $\delta\theta_{n+1}$ is a well-defined element of C^2 even though θ_{n+1} itself was purely formal. We have, therefore,

THEOREM 9. *If $\mathcal{C}$ is a* c.c.2., *and if an abelian element $\Phi_t = 1 + t\varphi_1 + t^2\varphi_2 + \cdots + t^n\varphi_n$ of $\operatorname{Aut}_n \mathcal{C}$ is written in the form $\Phi_t = \exp (t\theta_1 + t^2\theta_2 + \cdots)$, then $\delta\theta_i = 0$ for $i = 1, \cdots, n$ and $\delta\theta_{n+1}$ is a polynomial in the φ_i with integral coefficients, hence is a well-defined element of RZ^2.*

Note that $\delta\theta_{n+1}$ need not be the coboundary of an element of C^1.

Let $\pi_t = \Theta_t^{-1}(\Theta_t \smile \Theta_t)$ where $\Theta_t = 1 + t\theta_1 + t^2\theta_2 + \cdots$ for some $\theta_i \in C^1$, $i = 1, 2, \cdots$. Writing $\pi_t = 1 \smile 1 + tF_1 + t^1F_2 + \cdots$, one has $F_1 = \delta\theta_1$. It follows that, if an expression of the form $\pi_t = \exp t \sum \varphi_i \smile \psi_i$ is a trivial one-parameter family of deformations of a c.c.2. where the φ_i, ψ_i are contained in a commutative subring of C^1, then $\sum \varphi_i \smile \psi_i$ is a coboundary. There is a partial converse.

THEOREM 10. *Let $\mathcal{C}$ be a unital* c.c.2. *over $\mathbf{Q}$ and $\varphi_1, \cdots, \varphi_n, \psi_1, \cdots, \psi_n$ be commuting formal derivations of $\mathcal{C}$ such that $\sum \varphi_i \smile \psi_i = \delta\theta$ for some $\theta \in C^1$. If θ commutes with all the φ_i and ψ_j, then $\pi_t = \exp t \sum \varphi_i \smile \psi_i$ is a trivial one-parameter family of deformations of $\mathcal{C}$.*

PROOF. One may verify readily that, if θ commutes with every φ_i and

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ψ_j, then $(\operatorname{ad}\theta)^n(1 \smile 1) = \left(\sum \varphi_i \smile \psi_i\right)^n$. It follows that $\pi_t = e^{-t\theta}(e^{t\theta} \smile e^{t\theta})$, ending the proof.

For example, let φ, ψ be commuting derivations of $\mathcal{C}$. Then $\varphi \smile \psi + \psi \smile \varphi = -\delta(\varphi\psi)$. Since $\varphi\psi$ commutes with both φ and ψ, it follows that $\exp t(\varphi \smile \psi + \psi \smile \varphi)$ is trivial. Likewise, $\exp t(\varphi \smile \varphi)$ is trivial.

3. Dwork's lemma, Witt vectors

Let $x = (x_1, x_2, \cdots)$ be a sequence of commuting variables. A formal power series $f \in \mathbf{Q}[[x_1, x_2, \cdots]]$ is said to be integral at p if all the coefficients of f belong to the subring $\mathbf{Z}_p$ of $\mathbf{Q}$. Let the monomials μ in $x_1, x_2, \cdots$ be linearly ordered as follows. Set $\deg 1 = 0$, $\deg x_i = i$, and let the degree of a monomial be the sum of the degrees of its factors. Define $\mu < \mu'$ if either $\deg \mu < \deg \mu'$, or if $\deg \mu = \deg \mu'$ and μ precedes μ' in the lexicographic order. The sequence of monomials thus becomes $\mu_0 = 1$, $\mu_1 = x_1$, $\mu_2 = x_1^2$, $\mu_3 = x_2$, $\mu_4 = x_1^3$, $\mu_5 = x_1x_2$, $\cdots$. Multiplication respects this order. That is, if $\mu < \nu$ and $\mu' < \nu'$, then $\mu\mu' < \nu\nu'$. Moreover, $\mu_i\mu_j \geqq \mu_{i+j}$ with strict inequality if either $i, j > 1$. Every $f \in \mathbf{Q}[[x]]$ can be written uniquely in the form $f = a_0 + a_1\mu_1 + a_2\mu_2 + \cdots$ with $a_i \in \mathbf{Q}$. By definition, f is integral at p if $a_i \in \mathbf{Z}_p$ for all i, or equivalently, if $f \in \mathbf{Z}_p[[x]]$. One says that $f \equiv 0 \bmod p$ if $a_i \in p\mathbf{Z}_p$ for all i, and $f \equiv g \bmod p$ if $f - g \equiv 0 \bmod p$. If $f(x) = a_0 + a_1\mu_1 + a_2\mu_2 + \cdots$ is an element of $\mathbf{Q}[[x]]$, then by definition, $f(x^p) = a_0 + a_1\mu_1^p + a_2\mu_2^p + \cdots$.

LEMMA 2. (Dwork's lemma). *Let $f(x)$ be an element of $\mathbf{Q}[[x]]$ with leading coefficient 1. Then the necessary and sufficient condition that f be integral at p is that $f(x^p)/f(x)^p \equiv 1 \bmod p$.*

PROOF. The condition is clearly necessary. To prove it sufficient, write $f = 1 + a_1\mu_1 + a_2\mu_2 + \cdots$, and let n be the least integer such that $a_n \notin \mathbf{Z}_p$. Let I be the ideal of $\mathbf{Q}[[x]]$ consisting of all formal series of the form $b_{n+1}\mu_{n+1} + b_{n+2}\mu_{n+2} + \cdots$. Then

$$\begin{aligned} f(x)^p &\equiv (1 + a_1\mu_1 + \cdots + a_{n-1}\mu_{n-1})^p + pa_n\mu_n \\ &\equiv f(x^p) + pa_n\mu_n \qquad \bmod \left(I + p\mathbf{Z}_p[[x]]\right). \end{aligned}$$

Since $pa_n\mu_n \notin p\mathbf{Z}_p[[x]]$, it follows *a fortiori* that $f(x)^p/f(x^p) \not\equiv 1 \bmod p$.

We recall some basic facts from the Witt vector theory. Let a sequence of commuting variables $(x_0, x_1, \cdots)$ be given. (Note the change in indexing.) The prime p being fixed, set $w_n(x) = x_0^{p^n} + px_1^{p^{n-1}} + \cdots + p^nx_n$; this is the nth Witt polynomial. It is easy to see that x_i is a polynomial in $w_0(x), w_1(x), \cdots, w_i(x)$ with coefficients which are integral at every prime except p. If sequences (x) and (y) are given, then their Witt sum, $(x) +_w (y)$ is defined by

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setting $w_i((x) +_w (y)) = w_i(x) + w_i(y)$ for all i. Similarly, the Witt product $(x)\cdot_w (y)$ is defined by setting $w_i((x)\cdot_w (y)) = w_i(x)w_i(y)$. It is the case that the ith components of both $(x) +_w (y)$ and $(x)\cdot_w (y)$ are polynomials in $x_0, \cdots, x_i, y_0, \cdots, y_i$ with integer coefficients. We may therefore replace the variables by elements of any commutative associative ring R. The sequences of elements of R with Witt addition and multiplication then form a commutative associative ring $W(R)$, the Witt ring of R. It is a covariant functor of R.

Let commuting variables t and $(x_0, x_1, \cdots)$ be given. Then we define

$$e(t, (x)) = \exp \sum_{i=0}^{\infty} \frac{t^{p^i}}{p^i} w_i(x) .$$

It is immediate that $e(t, (x))$ satisfies the hypotheses of Dwork's lemma, so: *The series $e(t, (x))$ is integral at p.*

It follows that we may substitute for (x) a sequence of elements of an arbitrary commutative associative ring R. Observe that one has

$$e(t, (x))e(t, (y)) = e(t, (x) +_w (y)) . \tag{3.1}$$

It will be useful to state Dwork's lemma in exponential form. Let $(x) = (x_1, x_2, \cdots)$ be commuting variables and g be an element of $\mathbf{Q}[[x]]$ with leading coefficient zero. Then $\exp g(x)$ is integral at p if and only if $pg(x) \equiv g(x^p) \bmod p$. This form yields

LEMMA 3. *Let $t, x_0, x, \cdots$ be a sequence of commuting variables, and set $e(t, (x)) = 1 + ty_1 + t^2y_2 + \cdots$. Then for every i, x_i can be expressed as a polynomial in $y_1, y_2, \cdots, y_{p^i}$ with rational coefficients which are integral at p.*

PROOF. It is trivial that x_i is a polynomial $x_i(y_1, \cdots, y_{p^i})$; the problem is to prove its integrality. From the exponential form of Dwork's lemma one has

$$w_i(x(y))/p^{i-1} \equiv w_{i-1}(x(y^p))/p^{i-1} \qquad \bmod p , \tag{3.2}$$

where both sides of the congruence are considered as polynomials in the y's with rational coefficients. Since $x_0 = y_1$, it is clear that x_0 is integral. Assume that $x_0, x_1, \cdots, x_{i-1}$ are integral at p. Then (3.2) asserts that

$$\begin{aligned} &x_0(y)^{p^i} + px_1(y)^{p^{i-1}} + \cdots + p^i x_i(y) \\ &\quad \equiv x_0(y^p)^{p^{i-1}} + px_1(y^p)^{p^{i-2}} + \cdots + p^{i-1}x_{i-1}(y^p) \qquad \bmod p^i \end{aligned} \tag{3.3}$$

The integrality of x_j for $j < i$ implies that $x_j(y)^p \equiv x_j(y^p) \bmod p$, whence $x_j(y)^{p^{i-j}} \equiv x_j(y^p)^{i-j-1} \bmod p^{i-j}$, and therefore $p^j x_j(y)^{p^{i-j}} \equiv p^j x_j(y^p)^{i-j-1} \bmod p^i$.

It follows from (3.3) that $p^i x_i(y) \equiv 0 \bmod p^i$, whence $x_i(y)$ is indeed a polynomial in the y's which is integral at p. This completes the induction

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and the proof.

It is easy to show that, if $t, x_1, x_2, \cdots$ are commuting variables, then for every m prime to p, there exist polynomials $x_0^{(m)}, x_1^{(m)}, \cdots$ in the x's with coefficients integral at p, such that $1 + tx_1 + t^2x_2 + \cdots = \prod_{(m,p)=1} e(t^m, (x^{(m)}))$. Therefore, every abelian element of $\mathrm{Aut}_n \mathcal{C}$ is a product of elements of the form $e(t^m, \varphi^{(m)}))$ where the elements of C^1 in the sequences of $\varphi^{(m)}$ all commute, and where the series are all taken mod t^{n+1}. It follows from this, as one may show, then an element of $\mathrm{Aut}_\infty^a \mathcal{C}$ of the form $\Phi = 1 + t^l\varphi_l + t^{l+1}\varphi_{l+1} + \cdots$ has a factorization of the form $\Phi = \Phi'\Phi''$ where $\Phi', \Phi'' \in \mathrm{Aut}_\infty^a \mathcal{C}$, Φ' is a power series in t^l, and $\Phi'' = 1 + t^j\varphi_j + t^{j+1}\varphi''_{j+1} + \cdots$ with $j > l$. The proof is omitted for we shall not need these remarks.

4. Deformation formulas for composition complexes over $\mathbf{Z}_p$

If $(x) = (x_0, x_1, \cdots, x_n)$ is a finite sequence of commuting variables, then $w_i(x)$ is well-defined for $i \leqq n$. Set

$$e_n(t, (x)) = \exp \sum_{i=0}^{n} \frac{t^{p^i}}{p^i} w_i(x) .$$

This series is integral at p up to, but not including, terms of degree p^{n+1} or higher in t, and is tacitly taken mod $t^{p^{n+1}}$.

Since the exponential of a derivation is an automorphism, it follows from Theorem 9 that, if $\mathcal{C}$ is a c.c.2. over $\mathbf{Z}_p$ and $(\varphi) = (\varphi_0, \varphi_1, \cdots, \varphi_n)$ are commuting elements of C^1, *then* $e_n(t, (\varphi))$ *is in* $\mathrm{Aut}_{p^{n+1}-1} \mathcal{C}$ *if and only if* $\delta[(1/p^i)w_i(\varphi)] = 0$ *for* $i = 0, 1, \cdots, n$. *When that is so, then*

$$\delta\left[\frac{1}{p^{n+1}} w_n(\varphi^p)\right] = \delta\left[\frac{1}{p^{n+1}}\varphi_0^{p^{n+1}} + \frac{1}{p^n}\varphi_1^{p^n} + \cdots + \frac{1}{p}\varphi_n^p\right]$$

is integral at p, *and therefore a well-defined element of* RZ^2. If (φ) is an infinite sequence, then $e(t, (\varphi)) \in \mathrm{Aut}_\infty \mathcal{C}$ if and only if $\delta[(1/p^i)w_i(\varphi)] = 0$ for all i.

Thus, if $e(t, (\varphi)) = \exp[tw_0(\varphi) + (t^p/p)w_1(\varphi) + \cdots] \in \mathrm{Aut}_\infty \mathcal{C}$, then $\exp[tw_i(\varphi) + (t^p/p)w_{i+1}(\varphi) + \cdots] \in \mathrm{Aut}_\infty \mathcal{C}$ for all i.

Noting that, if x is a variable, then $\exp px \equiv 1 \bmod p$, and applying an argument analogous to that of Chapter I, § 3, one sees that, *if* $\mathcal{C}$ *has characteristic* p, *and* $(\varphi) = (\varphi_0, \varphi_1, \cdots, \varphi_n)$ *is a sequence of commuting elements of* C^1 *such that* $e_n(t, (\varphi)) \in \mathrm{Aut}_{p^{n+1}-1} \mathcal{C}$, *where the series* $e_n(t, (\varphi))$ *is understood* mod $t^{p^{n+1}}$, *then*

$$\Phi_i = \exp\left(tw_i(\varphi) + (t^p/p)w_{i+1}(\varphi) + \cdots + (t^{p^n}/p^n)w_{i+n}(\varphi)\right) \in \mathrm{Aut}_{p^{n+1}-1} \mathcal{C} .$$

Note that because the characteristic is p, this Φ_i can also be written in

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the form

$$\Phi_t = \exp\left[tw_0(\varphi^{p^i}) + (t^p/p)w_1(\varphi^{p^i}) + \cdots + (t^{p^n}/p^n)w_n(\varphi^{p^i})\right] = e_n\big(t, (\varphi^{p^i})\big) .$$

THEOREM 11. *Let $\mathcal{C}$ be an associative* c.c.3. *over $\mathbf{Z}_p$ and φ_i, ψ_i, $i = 0, 1, \cdots$ be commuting elements of C^1. Set $(\varphi) = (\varphi_0, \varphi_1, \cdots)$, $(\psi) = (\psi_0, \psi_1, \cdots)$, and suppose that $e\big(t, (\varphi)\big)$, $e\big(t, (\psi)\big)$ are both in $\mathrm{Aut}_\infty \mathcal{C}$. Set*

$$\pi_t = e\big(t; (\varphi), (\psi)\big) = \exp \sum_{i=0}^{\infty} \frac{t^{p^i}}{p^i} w_i(\varphi) \smile w_i(\psi) .$$

Then π_t is integral at p, hence an element of $C^2[[t]]$, and $\pi_t \circ \pi_t = 0$.

PROOF. We show first the integrality of π_t at p. Let S be the commutative subalgebra of C^1 generated by 1 and all the φ_i and ψ_i, and let $R = S \smile S$. Set $(\varphi \smile 1) = (\varphi_0 \smile 1, \varphi_1 \smile 1, \cdots)$ and $(1 \smile \psi) = (1 \smile \psi_0, 1 \smile \psi_1, \cdots)$. These are sequences of elements of the commutative ring R, and one has $w_n(\varphi) \smile w_n(\psi) = \big(w_n(\varphi) \smile 1\big)\big(1 \smile w_n(\psi)\big) = w(\varphi \smile 1) \cdot w_n(1 \smile \psi)$ for all n. Applying the Witt vector theory to the ring R, it follows that $w_n(\varphi) \smile w_n(\psi) = w_n(\mu)$ for some sequence $(\mu) = (\mu_0, \mu_1, \cdots)$ of elements of R. Thus $\pi_t = e\big(t, (\mu)\big)$ and therefore is integral at p.

To prove $\pi_t \circ \pi_t = 0$, observe that

$$\delta\left[\frac{1}{p^i} w_i(\varphi)\right] = \delta\left[\frac{1}{p^i} w_i(\psi)\right] = 0$$

for all i; as the $w_i(\varphi)$ and $w_i(\psi)$ all commute, we may apply Theorem 8 to conclude that π_t is, at least formally, a one-parameter family of deformations. Since it is actually meaningful over $\mathbf{Z}_p$, it follows that it is indeed a one-parameter family of deformations of $\mathcal{C}$.

In the case of characteristic p, the preceding theorem has the following refinement.

THEOREM 12. *Let $\mathcal{C}$ be an associative* c.c.3. *of characteristic p and φ_i, ψ_i, $i = 0, 1, \cdots, n$ be commuting elements of C^1. Set $(\varphi) = (\varphi_0, \varphi_1, \cdots, \varphi_n, 0, 0, \cdots)$ and $(\psi) = (\psi_0, \psi_1, \cdots, \psi_n, 0, 0, \cdots)$. Suppose that*

(i) *$\varphi_i^{p^{n+1}} = \psi_i^{p^{n+1}} = 0$ for $i = 0, 1, \cdots, n$ and*

(ii) *the polynomials*

$$e_n\big(t, (\varphi)\big) = \exp \sum_{i=0}^{n} \frac{t^{p^i}}{p^i} w_i(\varphi)$$

and $e_n\big(t, (\psi)\big)$, when considered mod $t^{p^{n+1}}$, *are both elements of* $\mathrm{Aut}_{p^{n+1}-1} \mathcal{C}$. *(Notice that* (i) *is equivalent to the assertion that $\big[e_n\big(t, (\varphi)\big)\big]^{p^{n+1}} = \big[e_n\big(t, (\psi)\big)\big]^{p^{n+1}} = 1$.) Let*

$$\pi_t = e\big(t; (\varphi), (\psi)\big) = \exp \sum_{i=0}^{2n} \frac{t^{p^i}}{p^i} w_i(\varphi) \smile w_i(\psi) .$$

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Then π_t is a polynomial in t integral at p and $\pi_t \circ \pi_t = 0$.

PROOF. The hypothesis (i) implies that π_t would not be changed if the summation in the formula defining π_t were extended to ∞ rather than to $2n$. That π_t is integral then follows as in the proof of Theorem 11. Further, if we knew that

$$\delta\left[\frac{1}{p^i} w_i(\varphi)\right] = \delta\left[\frac{1}{p^i} w_i(\psi)\right] = 0$$

for $i = 0, 1, \cdots, 2n$ rather than merely for $i = 0, 1, \cdots, n$, then it would follow just as in the proof of Theorem 11 that π_t is a non-parameter family of deformations. But observe now that if $(\varphi) = (\varphi_0, \varphi_1, \cdots)$ and $(\psi) = (\psi_0, \psi_1, \cdots)$ are any two sequences of commuting elements of C^1 such that $\varphi_i^{p^{n+1}} = 0$ for $i = 0, 1, \cdots, n$, then

$$\pi_t = \exp\left[\sum_{i=0}^{2n} \frac{t^{p^i}}{p^i} w_i(\varphi) \smile w_i(\psi)\right]$$

in fact does not depend on the φ_j and ψ_j with $j \geqq n + 1$. For in the expression in brackets these maps appear only in terms which have coefficients which are multiples of p. It follows, reasoning as in Chapter I, § 3, that in the presence of (i) we will have $\pi_t \circ \pi_t = 0$ whenever those equations

$$\delta\left[\frac{1}{p^i} w_i(\varphi)\right] = \delta\left[\frac{1}{p^i} w_i(\psi)\right] = 0$$

hold which do not involve any φ_j or ψ_j with $j > n$. But this is precisely what is assumed in (ii).

The following is an exact analog of Theorem 9 and the proof is therefore omitted.

THEOREM 13. *Let $\mathcal{C}$ be an associative* c.c.3. *over $\mathbf{Z}_p$, S be a commutative subring of C^1, and $(\varphi^{(j)}) = (\varphi_0^{(j)}, \varphi_1^{(j)}, \cdots)$, $(\psi^{(j)}) = (\psi_0^{(j)}, \psi_1^{(j)}, \cdots)$ $j = 1, \cdots, m$ be sequences of elements of S such that either* (i) *or* (ii) *holds.*

(i) *$e(t, (\varphi^{(j)}))$ and $e(t, (\psi^{(j)}))$ are in $\mathrm{Aut}_\infty \mathcal{C}$ for every j*

(ii) *$\mathcal{C}$ has characteristic p, and for every j there is an integer m_j such that setting $q_j = p^{m_j+1}$ we have*

(a) *$\varphi_i^{(j)}, \psi_i^{(j)}$ are nilpotent of index q_j for all j*

(b) *$\varphi_i^{(j)} = \psi_i^{(j)} = 0$ for $i > m_j$, and*

(c) *$e(t, (\varphi^{(j)}))$, $e(t, (\psi^{(j)}))$ are in $\mathrm{Aut}_{q_j-1}\mathcal{C}$ for all j. In case* (i), *set $m_j = \infty$ for all j. Then*

$$\pi_t = \prod_{j=1}^{m} e(t; (\varphi^{(j)}, \psi^{(j)})) = \exp \sum_{j=1}^{m} \sum_{i=0}^{2m_j} \frac{t^{p^i}}{p^i} w_i(\varphi^{(j)}) \smile w_i(\psi^{(j)}) \tag{4.1}$$

is a one-parameter family of deformations of $\mathcal{C}$.

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COROLLARY. *Fixing S, the one-parameter families of deformations given by* (4.1) *form an abelian group which will be denoted by $\pi(S)$.*

Theorem 11 yields a criterion for an element of $\pi(S)$ to be a trivial one-parameter family of deformations.

THEOREM 14. *Let $\mathcal{C}$ be an associative unital* c.c.3. *over $\mathbf{Z}_p$, S be a commutative subring of C^1, and $(\varphi) = (\varphi_0, \varphi_1, \cdots)$, $(\psi) = (\psi_0, \psi_1, \cdots)$ be sequences of elements of S. Suppose either that*

(i) $e(t, (\varphi)), e(t, (\psi)) \in \mathrm{Aut}_\infty \mathcal{C}$, *or that*

(ii) $\mathcal{C}$ *has characteristic p and there is an integer m such that φ_i, ψ_i are nilpotent of index p^{m+1} for all m, vanish for $i > m$, and*

$$e(t, (\varphi)), e(t, (\psi)) \in \mathrm{Aut}_{q-1} \mathcal{C}$$

where $q = p^{m+1} - 1$. Then

$$\Phi_t = \exp \sum \frac{t^{p^i}}{p^i} w_i(\varphi) w_i(\psi)$$

is integral at p, and

$$\Phi_t(\Phi_t^{-1} \smile \Phi_t^{-1}) = e(t; (\varphi), (\psi)) e(t; (\psi), (\varphi)) . \tag{4.2}$$

Thus $e(t; (\varphi), (\psi)) e(t; (\psi), (\varphi))$ is a trivial one-parameter family of deformations of $\mathcal{C}$.

PROOF. From the Witt vector theory, there is a sequence (θ) such that $\Phi_t = e(t, (\theta))$. This proves the integrality. As for equation (4.2), if f and g are derivations of $\mathcal{C}$, then $-\delta(fg) = f \smile g + g \smile f$. Since

$$\delta\left[\frac{1}{p^i} w_i(\varphi)\right] = \delta\left[\frac{1}{p^i} w_i(\psi)\right] = 0$$

for every i, it follows that

$$\delta\left[\frac{1}{p^i} w_i(\varphi) w_i(\psi)\right] = \frac{1}{p^i}[w_i(\varphi) \smile w_i(\psi) + w_i(\psi) \smile w_i(\varphi)] .$$

Applying Theorem 11 then gives the result.

Suppose now that π_t is a one-parameter family of deformations of an associative algebra A. Define π_t^{op} by setting $\pi_t^{\mathrm{op}}(a, b) = \pi_t(b, a)$. This is again an associative multiplication on $A[[t]]$. If A is commutative, then π_t^{op} is again a one-parameter family of deformations of A, whence the set of one-parameter families of deformations of a commutative associative algebra has an involution, op. It is easy to see that if S is a commutative subring of C^1, then op is an automorphism of $\pi(S)$. If $\alpha, \beta \in \pi(S)$ are both trivial families of deformations, then so is $\alpha\beta$. The trivial families therefore form a subgroup $\mathrm{triv}\, \pi(S)$, of $\pi(S)$.

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COROLLARY. *If A is a commutative associative algebra, S a commutative subring of $C^1(A, A)$, and $\alpha \in \pi(S)$, then $\alpha\alpha^{\mathrm{op}} \in \mathrm{triv}\ \pi(S)$.*

5. Automorphisms of algebras

When the composition complex under consideration is either of the form $C^*(X, k)$, X a finite simplicial complex, or $C^*(A, A)$, A a k-algebra, then certain special properties result from the fact that we can consider subcomplexes of X and subalgebras of A. We state these for the algebra case. The set of abelian elements $1 + t\varphi + \cdots$ of $\mathrm{Aut}_n A$ is denoted $\mathrm{Aut}_n^a A$, those φ which appear constitute $\mathrm{Der}_n^a A$.

THEOREM 15. 1. *Let A be an algebra over a ring k and $\Psi_t = 1 + t\psi_1 + t^2\psi_2 + \cdots + t^n\psi_n$ be an element of $\mathrm{Aut}_n^a A$, where it may be that $n = \infty$. Set $A^{(0)} = A$, and set $A^{(i)} = \{a \in A \mid \psi_1 a = \psi_2 a = \cdots = \psi_i a = 0\}$ for $i > 0$. Then $A^{(i)}$ is a subalgebra of A for every i, and the restriction of ψ_{i+1} to $A^{(i)}$ is a derivation of $A^{(i)}$.*

2. *Let A be an algebra over $\mathbf{Z}_p$, set $q = p^{n+1}$, and let an element of $\mathrm{Aut}_{q-1}^a A$ of the form $\Phi_t = e_n(t, (\varphi))$ be given. Set $A_i = \{a \in A \mid \varphi_0 a = \varphi_1 a = \cdots = \varphi_i a = 0\}$. Then A_i is a subalgebra of A for every i and the restriction of φ_{i+1} to A_i is a derivation of A_i.*

PROOF. (1) Suppose that $A^{(i)}$ is known to be a ring for some i; this is in particular the case for $i = 0$. Since the ψ_i commute, it follows that $\psi_j(A^{(i)}) \subset A^{(i)}$ for all i and j with $1 \leqq i, j \leqq n$. It follows that Ψ_t induces on $A^{(i)}$ an element of $\mathrm{Aut}_n^a A^{(i)}$. Let the restriction of Ψ_t to $A^{(i)}$ be denoted by $1 + t\psi_1' + \cdots + t^n\psi_n'$. Since by hypothesis $\psi_1' = \psi_2' = \cdots = \psi_i' = 0$, it follows that ψ_{i+1}' is a derivation (cf. I-(2.1)). The kernel of ψ_{i+1}', namely $A^{(i+1)}$, is therefore a ring, and the induction proceeds.

As for (2), suppose that A_i is a ring, and observe that the expansion of the restriction Φ_t' of Φ_t to A_i begins $\Phi_t' = 1 + t^{p^{i+1}}\varphi_{i+1}' + \cdots$. Therefore, the same argument applies.

Let $\pi_t = 1 \smile 1 + tF_1 + t^2F_2 + \cdots$ be a one-parameter family of deformations of an associative algebra A, and denote by A_t the $k[[t]]$-algebra $A[[t]]$ with the multiplication π_t. Let $\Psi_u = 1 + u\psi_1 + u^2\psi_2 + \cdots + u^n\psi_n$ be an element of $\mathrm{Aut}_n A$ with the property that $\Psi_u F_i(a, b) = F_i(\Psi_u a, \Psi_u b)$ for all i and all $a, b \in A$. (The parameter has been changed to u to avoid confusion with the parameter in π_t.) Then clearly $\Psi_u \in \mathrm{Aut}_n A_t$. It follows, for example, that ψ_1 is a derivation of A_t. If $\Psi_u \in \mathrm{Aut}_n^a A$, then we may apply the preceding theorem to A_t.

The explicit deformation formulas provide an instance where this remark is useful. For example, suppose that $\varphi_0, \varphi_1, \cdots, \psi_0, \psi_1, \cdots$ are commuting

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elements of $C^1(A, A)$, A a $\mathbf{Z}_p$-algebra, such that $\Phi_t = e(t, (\varphi))$ and $\Psi_t = e(t, (\psi))$ are one-parameter families of automorphisms of A. If $\pi_t = e(t; (\varphi), (\psi))$ is written in the form $1 \smile 1 + tF_1 + t^2F_2 + \cdots$, then each F_i is a sum of terms of the form $f_i \smile f_i'$, where f_i is a polynomial in the φ's and f_i' a polynomial in the ψ's. Since the φ's and ψ's all commute, one has immediately that $\Phi_t(f_i \smile f_i') = \Phi_t f_i \smile \Phi_t f_i'$. This yields the required condition. The same observation holds for Ψ_t', so it follows that Φ_u, Ψ_u are both one-parameter families of automorphisms of the algebra A_t defined by the multiplication $\pi_t = e(t; (\varphi), (\psi))$.

THEOREM 16. *Let A be a commutative algebra of characteristic p.*

(1) *If $\Psi_t = 1 + t\psi_1 + \cdots + t^n\psi_n$ is an element of $\mathrm{Aut}_n A$, then $(\psi_i a)^p = \psi_{pi}a^p$ for all $a \in A$ and all i such that $pi \leqq n$, while $\psi_j a^p = 0$ if $p \nmid j$.*

(2) *If $\varphi_0, \varphi_1, \cdots, \varphi_n$ are commuting elements of $C^1(A, A)$ such that $e(t, (\varphi))$ is in $\mathrm{Aut}_{p^{n+1}-1}A$, and if $\varphi_0 a = \varphi_1 a = \cdots = \varphi_{i-1}a = 0$ for some $a \in A$, then $(\varphi_i a)^p = \varphi_{i+1}a^p$.*

PROOF. (1). Since Ψ_t is an automorphism mod t^{n+1}, one has $\Psi_t a^p \equiv (\Psi_t a)^p$ mod t^{n+1}. Comparing coefficients of powers of t gives the result.

(2) With the notation of Theorem 15, one has $a \in A_{i-1}$. Since the restriction of φ_i to A_{i-1} is a derivation, it follows that $\varphi_i a^p = 0$. Comparing powers of t in the congruence $[e(t, (\varphi))a]^p \equiv e(t, (\varphi))a^p$ mod $t^{p^{n+1}}$ then gives the result.

CHAPTER III. DEFORMATION OF INSEPARABLE FIELDS

1. Purely inseparable fields

An extension K of a field k of characteristic p is purely inseparable if, for every $a \in K$, there is an n such that $a^{p^n} \in k$. The least such n is the exponent of a. The supremum of the exponents of all $a \in K$ is called the exponent of K. For fields of exponent 1, there is a "Galois theory" due initially to Jacobson (cf. [11], [5], and [6]) which associates to every field L with $K \supset L \supset k$ the set $\mathrm{Der}\, K/L$ of derivations of K which vanish on L. The necessary and sufficient condition that a subset $V \subset \mathrm{Der}\, K$ be of the form $\mathrm{Der}\, K/L$ for some L is that V be a vector space over K, be restricted, i.e., have the property that $\varphi \in V$ implies $\varphi^p \in V$, and be closed in the Krull-type topology introduced in [6] (a vacuous condition if $[K: k] < \infty$).

If K is purely inseparable over k, if $a \in K$ has exponent n, and $\varphi \in \mathrm{Der}_{n+1}K$, then $\varphi a = 0$. For let $\Phi_t = 1 + t\varphi + \cdots + t^q\varphi_q$ be an element of $\mathrm{Aut}_q K$ with $q = p^n$; we tacitly set $t^{q+1} = 0$. Then $\Phi_t a^q = a^q$, since $a^q \in k$, whence $\varphi_i a^q = 0$ for all i. But $\Phi_t a^q = (\Phi_t a)^q = a^q + t^q(\varphi a)^q$, whence $\varphi a = 0$. Therefore, if K is purely inseparable over k, then $\bigcap_{n=0}^{\infty} \mathrm{Der}_n K = 0$. The same argument

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shows that if K has exponent e then, in fact, $\mathrm{Der}_e K = 0$. (On the other hand, one can have $\mathrm{Der}_e K = 0$ for K of exponent greater than e.) It follows that, if K has exponent e, then every $\varphi \in \mathrm{Der}\, K$ is obstructed at no later than the pq^{th} place, where $q = p^e$, whence the direct limit of the groups $\mathrm{Aut}_m K$, as taken in Chapter I, § 6, is isomorphic to $\mathrm{Aut}_{pq-1} K$.

An $a \in K$ is fixed by an element $\Phi_t = 1 + t\varphi + \cdots + t^m\varphi_m$ of $\mathrm{Aut}_m K$ if $\Phi_t a = a$; i.e., if $\varphi_i a = 0$, $i = 1, \cdots, m$. The elements fixed by a subgroup G of $\lim \mathrm{dir}\, \mathrm{Aut}_m K$ form a field called the fixed field of G.

We shall call a purely inseparable extension K of k *modular* over k if there is a p-basis X for K such that $K = \bigotimes_{x \in X} k(x)$; such a basis is called a modular basis. (The tensor product is taken relative to k.) Thus, a modular extension is here defined to be a possibly infinite tensor product of simple purely inseparable extensions. (This is at slight variance with other definitions (cf. a forthcoming paper of Sweedler [14]). An extension of exponent 1 is always modular, but it is easy to show by example that, if K is a modular (even finite) extension of k of exponent greater than 1, then there generally exist fields L with $K \supset L \supset k$ such that K is not modular over L, L is not modular over k, or both.

If $k(x)$ is a simple purely inseparable extension of k of exponent n (i.e., $x^{p^n} \in k$, $x^{p^{n-1}} \notin k$), then $k(x)$ is isomorphic to $k[T]/(T^{p^n} - x^{p^n})$, where T is a variable. Setting $q = p^n$, it follows that, if a is any element of $k(x)$, then we can define an element of $\mathrm{Aut}_{q-1} k(x)$ by setting $\varphi_t x = x + ta$. More generally, one obtains an element of $\mathrm{Aut}_{iq-1} k(x)$ by setting $\Phi_t x = x + t^i a$. It follows that, if $K = \bigotimes_{x \in X} k(x)$ is modular and of finite exponent e, then one can define an element $\Phi_t \in \mathrm{Aut}_{p^e-1} K$ as follows. If $x \in X$ has exponent n, set $\Phi_t x = x + t^{p^e-n} a_x$ where a_x is an arbitrary element of K varying with x. It is easy to verify that, if one chooses $a_x = x^{p+1}$, then the fixed field of Φ_t is reduced to k. *A fortiori*, the fixed field of $\mathrm{Aut}_{p^e-1} K$ is reduced to k. By the same technique, one can show more generally that, if K is modular over k but not necessarily of finite exponent, then there is an element of $\lim \mathrm{dir}\, \mathrm{Aut}_m K$ whose fixed field is reduced to k.

Weisfeld has shown in his thesis [16] that K is modular and of finite exponent over k if and only if, for some m, there is an element $\Phi_t \in \mathrm{Aut}_m K$ whose fixed field is reduced to k. More recently, Sweedler [14] has shown that *if K is purely inseparable and of finite exponent e over k, then K is modular if and only if the fixed field of all* $\mathrm{Aut}_{p^e-1} K$ *is reduced to k.*

From this theorem of Sweedler, it follows that, given any purely inseparable extension of finite exponent e, $K \supset k$, there is a smallest field F with $K \supset F \supset k$ such that K is modular over F. Indeed, F is just the fixed field

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for $\lim \operatorname{dir} \operatorname{Aut} K = \operatorname{Aut}_{q-1} K$ where $q = p^e$. Observe, incidentally, that if $\Phi_t = 1 + t\varphi_1 + \cdots + t^m\varphi_m$ is an element of $\operatorname{Aut}_m K$ for some m, then every φ_i is linear over F. For if $c \in F$, then $\Phi_t(ca) = \Phi_t c \cdot \Phi_t a = c\Phi_t a$. (Tacitly, $t^{m+1} = 0$.)

Keeping the foregoing notation, let X be a modular basis for K over F, and write $X = X_e \cup X'$ where $X_e = \{x \in X \mid \text{exponent of } x = e\}$. Since K has exponent e, one has $K = k(X_e) \otimes_k F(X')$. If $\varphi \in \operatorname{Der}_{e-1} K$, then φ vanishes on $F(X')$ and on X_e^p. Conversely, if φ vanishes on $F(X' \cup X_e^p)$ then one can define an element of of $\operatorname{Aut}_{q-1} K$, $q = p^e$, by setting $\Phi_t x = x + t\varphi(x)$ for all $x \in X$. Thus $\varphi \in \operatorname{Der}_{e-1} K$ if and only if φ vanishes on $F(X' \cup X_e^p)$. It follows that $\operatorname{Der}_{e-1} K$ is closed in the Krull topology, and that the field associated to it by the Jacobson Galois theory is just $F(X' \cup X_e^p)$. In particular, the latter field does not depend on the choice of modular basis X of K over F. More generally, if we set $X = X_1 \cup X_2 \cup \cdots \cup X_e$ where $X_i = \{x \in X \mid \text{exponent of } x = i\}$, then $\operatorname{Der}_i K$ is closed for all i, and the field associated to it is $F(X_1 \cup \cdots \cup X_i \cup X_{i+1}^p \cup \cdots \cup X_e^p)$.

2. A deformation of a division ring is a division ring

Let A be an algebra with a decreasing filtration over the coefficient ring k. That is, $A \supset \cdots \supset F_i A \supset F_{i+1} A \supset \cdots$ where the $F_i A$ are k-submodules such that $F_i A \cdot F_j A \supset F_{i+j} A$. The index i may run through all the integers $\mathbf{Z}$. One calls A complete if

(1) $\bigcup F_i A = A$;

(2) $\bigcap F_i A = 0$;

(3) A is complete in the topology obtained by taking the $F_i A$ as open neighborhoods of 0.

Set $\operatorname{gr}_i A = F_i A / F_{i+1} A$ and let gr_i denote the canonical k-module epimorphism $F_i A \to \operatorname{gr}_i A$. Then $\operatorname{gr} A = \bigoplus_i \operatorname{gr}_i A$ acquires a natural ring structure so, if $a \in \operatorname{gr}_i A$ and $b \in \operatorname{gr}_j A$, then $ab = \operatorname{gr}_{i+j}(\alpha\beta)$ where α is any representative of a in $F_i A$, and β is any representative of b. This is the associated graded ring of A. It is filtered by setting $F_n \operatorname{gr} A = \bigoplus_{i \geq n} \operatorname{gr}_i A$; the completion in this filtration is called the completed graded ring, $\operatorname{cgr} A$ of A. If A is complete, then A and $\operatorname{cgr} A$ are homeomorphic as topological spaces by a homeomorphism which is not canonical. One of the main results of [7] was that a complete filtered ring may be viewed as a deformation of its completed associated graded ring.

THEOREM 17. *Let A be a complete filtered associative ring.*

1. *If* $\operatorname{cgr} A$ *has no left (right) zero divisors, then neither has A.*
2. *If* $\operatorname{cgr} A$ *has a unit element, then so has A.*

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3. *If A has a unit element, and if $a \in F_iA$ is an element such that* $\mathrm{gr}_i a$ *has a left (right) inverse, then a has a left (right) inverse.*

PROOF. Assertion 1 is trivial. In 2, it is sufficient to show that there is an idempotent $e \in F_0A$ such that $\mathrm{gr}_0 e = 1$. For $e(ea - a) = (ae - a)e = 0$, whence as e is neither a left nor a right zero divisor, we have $ea = a = ae$. Now to contruct such an e, let e_1 be any element in F_0A such that $\mathrm{gr}_0 e_1 = 1$, and define e_n inductively, $n = 1, 2, \cdots$ by setting $e_{n+1} = 2e_n - e_n^2$. Now $e_1 - e_1^2 \in F_1A$.

If we assume that $e_n - e_n^2 \in F_nA$, then as one may readily verify,

$$e_{n+1} - e_{n+1}^2 = 2[(e_n - e_n^2) - e_n(e_n - e_n^2)] - [e_n - e_n^2]^2 \in F_{n+1}A .$$

It follows that $e_{n+1} - e_n = e_n - e_n^2$ is in F_nA for all n, so the sequence $e_1, e_2, \cdots$ converges to the desired idempotent e.

As for 3, observe that, if $b \in F_1A$, then $(1 + b)^{-1} = 1 - b + b^2 - \cdots$, a convergent series. If a is an arbitrary element of F_iA such that gr_i has a left inverse, then that left inverse is of the form $\mathrm{gr}_{-i} a'$ for some $a' \in F_{-i}A$. One then has $\mathrm{gr}_0 a'a = 1$, whence $a'a$ is of the form $1 + b$ for some $b \in F_1A$. It is therefore invertible, and the desired left inverse of a is then $(a'a)^{-1}a'$. This ends the proof.

COROLLARY. *Let A be a complete filtered algebra such that* $\mathrm{cgr}\, A$ *is a division algebra. Then A is a division algebra.*

Suppose now that A is a division algebra with no filtration over a field k, and $\Pi_t = 1 \frown 1 + tF_1 + t^2F_2 + \cdots$ be a one-parameter family of deformations of A. Let A_t denote the $k((t))$-algebra whose underlying vector space is $A((t))$ but with multiplication given by Π_t. Then A_t is filtered by setting $F_nA_t = \{\sum_{i \geq n}^{\infty} a_i t^i\}$, and with this filtration, $\mathrm{cgr}\, A_t$ is just $A((t))$ with the usual multiplication. The latter is obviously a division ring over $k((t))$, whence so is A_t. Note that A may be a field, but A_t will be a non-commutative division ring unless $F_i(a, b) = F_i(b, a)$ for all $a, b \in A$.

A one-parameter family $\Pi_t = 1 \frown 1 + tF_1 + \cdots$ of deformations of A is *locally finite* if for all $a, b \in A$ there is an n such that $F_i(a, b) = 0$ whenever $i > n$. A locally finite deformation Π_t defines a new multiplication on the $k[t]$-module $A[t]$. If A is a division algebra, then $A[t]$, with the Π_t multiplication, can be embedded in the division algebra A_t; the sub-division ring generated by $A[t]$ is then a division algebra D over $k(t)$. Note that $A[t]$ with the Π_t multiplication is both left and right noetherian (cf. [12, p. 163 ff]) and therefore has both a left and a right quotient division ring (*loc. cit.*, p. 165); these coincide with each other and with D. Suppose now that A is a finite-dimensional division algebra over the field k. Then $A(t) = k(t) \otimes_k A$, with

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the Π_t multiplication is a finite dimensional algebra without zero divisors over $k(t)$, and hence $A(t) = D$.

3. Deformation of $\mathbf{Z}[x_0, x_1, \cdots]$

Let $\mathbf{Z}$ denote the ring of integers, and $(x) = (x_0, x_1, \cdots)$ be commuting variables. By an abuse of notation, the ring $R = \mathbf{Z}[x_0, x_1, \cdots]$ will sometimes be denoted simply $\mathbf{Z}[x]$. We write $\mathbf{Z}[x_0, x_1, \cdots, x_n] = R_n$. As before, p denotes a fixed prime and we set $w_i(x) = x_0^{p^i} + px_1^{p^{i-1}} + \cdots + p^i x_i$, the ith Witt polynomial. If t is a variable and $y_0, y_1, \cdots$ are arbitrary elements of $R[t]$, then there is a unique automorphism Φ_t of $R[t]$ defined by setting $\Phi_t x_i = x_i + ty_i$. We write $(\Phi_t x) = (\Phi_t x_0, \Phi_t x_1, \cdots)$, and now fix once and for all the following definition: $(\Gamma_t x)$ is the sequence defined by setting $w_i(\Gamma_t x) = w_i(x) + t^{p^i}$.

LEMMA 4. *For all i, $\Gamma_t x_i \in R_i[t]$, and the automorphism of $R[t]$ defined by $(x) \mapsto (\Gamma_t x)$ is an abelian one-parameter family of automorphisms of R carrying $R_i[t]$ into itself.*

PROOF. Set $\mathbf{t} = (t, 0, 0, \cdots)$. Then $(\Gamma_t x)$ is just the Witt sum of the sequences (x) and $\mathbf{t}$, proving that $\Gamma_t x_i$ is a polynomial in $x_0, x_1, \cdots, x_i$ and t with integer coefficients. It is obvious that $\Gamma_t x_i$ is of the form $x_i + ty_i$ with $x_i \in R_i[t]$. Also, if t and u are variables, then it is obvious from the definition that $(\Gamma_t \Gamma_u x) = (\Gamma_u \Gamma_t x)$, which proves that the family Γ_t is abelian.

If φ is a k-module endomorphism of an algebra A and $a \in A$, then we may sometimes write $\langle \varphi, a \rangle$ instead of $\varphi(a)$.

LEMMA 5. *The family Γ_t is of the form*

$$\Gamma_t = e\big(t, (\gamma)\big) = \exp \sum_{i=0}^{\infty} \frac{t^{p^i}}{p^i} w_i(\gamma) ,$$

where $(\gamma) = (\gamma_0, \gamma_1, \cdots)$ is a sequence of commuting additive maps of R into itself such that $\gamma_i(R_i) \subset R_i$. One has

$$\langle w_i(\gamma), w_j(x) \rangle = \delta_{ij} p^i$$

where δ_{ij} is the Kronecker delta.

PROOF. Express $\log \Gamma_t$ as a formal power series in t: $\log \Gamma_t = (\Gamma_t - 1) - (\Gamma_t - 1)^2/2 + \cdots$. Denoting the rationals by $\mathbf{Q}$, this can be written in the form $\log \Gamma_t = t\theta_1 + t^2\theta_2 + \cdots$, where by Theorem 16 each θ_i is a derivation of $\mathbf{Q}[x]$ into itself. It is easy to see that θ_i carries $\mathbf{Q}[x_0, \cdots, x_i]$ into itself.

Now $\langle \Gamma_t - 1, w_i(x) \rangle = t^{p^i}$, which implies that $\langle (\Gamma_t - 1)^2, w_i(x) \rangle = 0$. Therefore,

$$\langle \log \Gamma_t, w_i(x) \rangle = t^{p^i} . \tag{3.1}$$

It follows that, if j is not a power of p, then we have $\langle \theta_j, w_i(x) \rangle = 0$ for all

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i. But the $w_i(x)$ generate $\mathbf{Q}[x]$ over $\mathbf{Q}$. Therefore, $\theta_j = 0$ unless j is a power of p. We may now define a sequence of additive maps $\gamma_i: \mathbf{Q}[x] \to \mathbf{Q}[x]$ by setting $w_i(\gamma) = p^i\theta_{p^i}$. It then follows that $\Gamma_t = e(t, (\gamma))$. But Γ_t is integral, that is, it carries $\mathbf{Z}[x] = R$ into itself. Therefore, every γ_i is also integral. The fact that $\gamma_i(R_i) \subset R_i$ is evident. Finally, from Equation (3.1), we have $\langle \theta_{p^i}, w_j(x)\rangle = \delta_{ij}$, whence $\langle w_i(\gamma), w_j(x)\rangle = \delta_{ij}p^i$.

COROLLARY. $\langle w_i(\gamma), x_j\rangle = 0$ *for* $j < i$.

PROOF. One can write x_j as a polynomial in $w_0(x), \cdots, w_j(x)$ with rational coefficients which are integral at every prime but p. It follows that any derivation vanishing on $w_0(\gamma), \cdots, w_j(\gamma)$ likewise vanishes on x_j. The $w_i(\gamma)$ are derivations by Chapter II, § 3.

It follows from Chapter II, § 4 that $\exp \sum_{i=0}^{\infty} (t^{p^i}/p^i)w_{i+1}(\gamma)$ is again a one-parameter family of automorphisms of R. Therefore, denoting as before by $\mathbf{Z}_p$ the localization of $\mathbf{Z}$ at the prime p, there is, by Theorem 11, a one-parameter family of deformations of $\mathbf{Z}_p[x_0, x_1, \cdots]$ defined by setting

$$\Pi_t = \exp \sum_{i=0}^{\infty} \frac{t^{p^i}}{p^i} w_i(\gamma) \frown w_{i+1}(\gamma) .$$

We show next that Π_t is integral, that is, if $y, z \in Z[x]$, then $\Pi_t(y, z) \in \mathbf{Z}[x]$. It will follow that Π_t may be viewed as a one-parameter family of deformations of $\mathbf{Z}[x]$. To this end, set $w_i(x) = \xi_i$ and $p^{-i}w_i(\gamma) = \alpha_i$. It follows that the α_i are derivations of $\mathbf{Z}[\xi] = \mathbf{Z}[\xi_0, \xi_1, \cdots]$, and one has $\langle \alpha_i, \xi_j\rangle = \delta_{ij}$. The latter relation implies that $\alpha_i^m/m!$ is integral on $\mathbf{Z}[x]$. For if μ is a monomial in $\xi_0, \xi_1, \cdots$, write $\mu = \xi_i^n\mu_i$, where $\xi_i \nmid \mu_i$. Then $\langle \alpha_i^m/m!, \mu\rangle = \binom{n}{m}\xi_i^{n-m}\mu_i$. This in turn implies that

$$\Pi_t = \exp \sum_{i=0}^{\infty} (t^{p^i}/p^i)\alpha_i \frown \alpha_{i+1}$$

is integral on $\mathbf{Z}[\xi_0, \xi_1, \cdots]$. But for every m, it is the case that x_m can be expressed as a polynomial in $\xi_0 = w_0(x), \cdots, \xi_m = w_m(x)$, with coefficients which are integral at every prime except p. It follows that any integral polynomial in $x_0, x_1, \cdots$ can also be expressed as a polynomial in $\xi_0, \xi_1, \cdots$ with coefficients which are integral at every prime except possibly p. Therefore, if $y, z \in \mathbf{Z}[x]$, then $\Pi_t(y, z)$ can be expressed as a polynomial in $x_0, x_1, \cdots$ with rational coefficients which are integral at every prime except possibly p. But Π_t has already been shown to be integral at p. Summarizing, we have proven the following

THEOREM 18. *Fix a prime* p *and define a one-parameter family* Γ_t *of automorphisms of* $\mathbf{Z}[x] = \mathbf{Z}[x_0, x_1, \cdots]$ *by setting* $w_i(\Gamma_t x) = w_i(x) + t^{p^i}$. *Then* Γ_t *is of the form* $e(t, (\gamma))$ *and*

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$$\Pi_t = \exp \sum_{i=0}^{\infty} \frac{t^{p^i}}{p^i} w_i(\gamma) \smile w_{i+1}(\gamma)$$

is a one-parameter family of deformations of $\mathbf{Z}[x]$.

It is easy to see that the deformation Π_t *has the following properties, which in fact characterize it completely.*

1. $\Pi_t(1, y) = \Pi_t(y, 1) = y$ *for all* $y \in \mathbf{Z}[x]$, i.e., 1 *remains the unit element after deformation.*

2. $\Pi_t\big(w_i(x), w_j(x)\big) = w_i(x)w_j(x)$ *if* $j \neq i + 1$, *while*

3. $\Pi_t\big(w_i(x), w_{i+1}(x)\big) = w_i(x)w_{i+1}(x) + p^{i+1}t^{p^i}$.

Therefore, writing $\Pi_t(y, z) - \Pi_t(z, y) = [y, z]_t$, *one has* $[w_i(x), w_j(x)]_t = 0$ *if* $j \neq i + 1, i - 1$, *while* $[w_i(x), w_{i+1}(x)]_t = p^{i+1}t^{p^i}$.

We have seen that, for all n, $R_n[[t]]$ is stable under Γ_t. It will be shown that in fact $R_n[t]$ is stable and the restriction of Γ_t to $R_n[t]$ is a polynomial in t.

LEMMA 6. *For every* i, *we have* $\gamma_0 x_i \in R_{i-1} = \mathbf{Z}[x_0, \cdots, x_{i-1}]$.

PROOF. Since $\gamma_0 x_0 = \langle w_0(\gamma), w_0(x)\rangle = 1$, the assertion is true for $i = 0$. For larger i, we have $\langle \gamma_0, x_0^{p^i} + px_1^{p^{i-1}} + \cdots + p^i x_i\rangle = 0$, which implies that $p^i(\gamma_0 x_i) \in \gamma_0\big(\mathbf{Z}[x_0, \cdots, x_{i-1}]\big)$. Making the inductive hypothesis that the assertion is true for $i - 1$, it follows that $p^i(\gamma_0 x_i) \in \mathbf{Z}[x_0, \cdots, x_{i-1}]$. Since γ_0 is integral, it follows that $\gamma_0 x_0 \in \mathbf{Z}[x_0, \cdots, x_{i-1}]$, completing the induction and the proof.

By virtue of the lemma, we may set $\gamma_0 x_i = g_i(x_0, x_1, \cdots, x_{i-1})$, where the g_i are certain polynomials; this notation will be retained.

The ring R becomes graded by setting $\deg x_i = p^i$, $i = 0, 1, \cdots$. With this grading, each $w_i(x)$ is homegeneous of degree p^i. If G is a graded group and f an endomorphism of G, then f is homogeneous of degree r, by definition, if $\deg(fa) = (\deg a) + r$ whenever a is a homogeneous element of G. Equation 3.1 implies that $\log \Gamma_t$ is a homogeneous mapping of $R[t]$ into itself of degree zero. This in turn implies, by induction on i, that each γ_i is homogeneous of degree $-p^i$. In particular, we have

$$\gamma_i^{p^j} x_l \in Z \qquad \text{if} \quad i + j = l \tag{3.2}$$

$$\gamma_i^{p^j} x_l = 0 \qquad \text{if} \quad i + j > l\,. \tag{3.3}$$

The stability of $R_n[t]$ under Γ_t follows from Equation 3.3. Our information can be summarized in

THEOREM 19. 1. $\Gamma_t\big(R_n[t]\big) \subset R_n[t]$ *for all* n.

2. *If* $y \in R_n[t]$, *then* $\Gamma_t^{p^{n+1}} y = y$. *Denoting the restrictions of* Γ_t *and* γ_i *to* $R_n[t]$ *by* ${}_n\Gamma_t$ *and* ${}_n\gamma_i$, *respectively, one has*

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$$ {}_n\Gamma_t^{p^{n+1}} = 1 \qquad \textit{and} \qquad {}_n\gamma_i^{p^{n+1}} = 0 \ . $$

3. ${}_n\Gamma_t$ *is a polynomial in* t.

4. $\Pi_t(R_n[t], R_n[t]) \subset R_n[t]$. *The restriction* ${}_n\Pi_t$ *of* Π_t *to* $R_n[t] \otimes_{\mathbf{Z}[t]} R_n[t]$ *is a polynomial in* t *for every finite* n.

The following lemma gives some information about the integer appearing in (3.2).

LEMMA 7. *For all* i, $\gamma_0^{p^i} x_i \equiv 1 \mod p$.

PROOF. The proposition is true for $i = 0$. We make the inductive assumption that it is true for all integers less than i. Since $\gamma_0 x_0 = 1$ and $\langle \gamma_0, w_i(x) \rangle = 0$ for $i > 0$, we have

$$ x_0^{p^{i}-1} + x_1^{p^{i-1}-1}(\gamma_0 x_1) + \cdots + x_{i-1}^{p-1}(\gamma_0 x_{i-1}) + (\gamma_0 x_i) = 0 \ . $$

By Equation (3.3), the inductive hypothesis, and the fact that $\gamma_0^{p^{i-1}}$ is a derivation modulo p, we have

$$ (p-1) x_{i-1}^{p-2} \gamma_0 x_{i-1} + \gamma_0^{1+p^{i-1}} x_i \equiv 0 \mod p \ . $$

Applying $\gamma_0^{p^{i-1}}$ again $p-2$ times, we have

$$ (p-1)! \gamma_0 x_{i-1} + \gamma_0^{1+(p-1)p^{i-1}} x_i \equiv 0 \mod p \ , $$

or

$$ \gamma_0^{1+(p-1)p^{i-1}} x_i \equiv \gamma_0 x_{i-1} \mod p \ . $$

Applying $\gamma_0^{p^{i-1-1}}$ completes the induction.

In an arbitrary ring A, we shall write $a \equiv 0 \bmod m$ to indicate that a is in the ideal mA consisting of elements divisible by some integer m.

THEOREM 20. *Let an integer* $n > 0$ *and elements* $y, z \in \mathbf{Z}[x]$ *be given. Then for* $i = 0, 1, \cdots, n$, *one has*

$$ \gamma_i(y^{p^{n+1}} z) \equiv y^{p^{n+1}} (\gamma_i z) \mod p \ . $$

PROOF. One has $\Gamma_t(y^{p^{n+1}} z) = (\Gamma_t y)^{p^{n+1}} (\Gamma_t z)$. Writing $\Gamma_t y = y + ty'$ where $y' \in \mathbf{Z}[x, t]$, one has

$$ \Gamma_t(y^{p^{n+1}} z) \equiv y^{p^{n+1}} \Gamma_t z \mod (p, t^{p^{n+1}}) \ , \tag{3.4} $$

where $(p, t^{p^{n+1}})$ is the ideal generated by p and $t^{p^{n+1}}$. If we write $\Gamma_t = 1 + t\varphi_1 + t^2\varphi_2 + \cdots$, then it follows by comparing the powers of t in (3.4) that

$$ \varphi_i(y^{p^{n+1}} z) \equiv y^{p^{n+2}} \varphi_i(z) \mod p $$

for $i = 1, 2, \cdots, p^{n+1} - 1$. By Theorem 12, γ_i is a polynomial in $\varphi_1, \cdots, \varphi_{p^i}$ with coefficients which are integral at p. This ends the proof.

COROLLARY. $\gamma_i y^{p^{n+1}} \equiv 0 \bmod p$ *for* $i = 0, 1, \cdots, n$.

PROOF. Set $z = 1$ and observe that $\gamma_i 1 = 0$ for all i.

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4. Nilpotent derivations of a field

LEMMA 8. *Let K be a field of characteristic p, φ be a derivation of K which is nilpotent of index N* (i.e., $\varphi^N = 0, \varphi^{N-1} \neq 0$) *and L be the constant field of φ. Then K is a purely inseparable extension of L with $[K : L] = N$. In particular, $N = p^{m+1}$ for some m. If $(\xi) = (\xi_0, \xi_1, \cdots, \xi_m)$ is a sequence of elements of K such that for every $i = 0, 1, \cdots, m$, we have $\varphi^{p^i}\xi_i \neq 0, \varphi^{p^{i+1}}\xi_i = 0$, then (ξ) is a p-basis for K over L.*

PROOF. Consider K as a vector space over L. The null space of φ then has dimension 1, being spanned by the unit element 1 of K. Therefore, $\dim_L K = N$. Since $a^p \in L$ for all $a \in K$, it follows that K is purely inseparable, of exponent one, and of degree $N = p^{m+1}$ over L. Now, given the sequence $(\xi) = (\xi_0, \cdots, \xi_m)$, set $K_i = L(\xi_0, \xi_1, \cdots, \xi_i)$. Since $\varphi^{p^{i+1}}$ is a derivation annihilating the generators of K_i, it follows that it annihilates all of K_i. But it does not annihilate ξ_{i+1}, so $\xi_{i+1} \notin K_i$, proving that (ξ) is a p-basis.

THEOREM 21. *Let K be a purely inseparable extension of a field k of characteristic p and φ be a derivation of K over k with the following property: There is an integer n such that $\varphi \in \mathrm{Der}_n K/k$ and $\varphi^{p^{n+1}} = 0$. Then $\varphi \in \mathrm{Der}_n^a K/k$.*

PROOF. Let φ have nil-index p^{m+1}, where, by hypothesis, $m \leqq n$. If L is the constant field for φ, then by the preceding lemma, we have $[K : L] = p^{m+1}$. If $(\xi) = (\xi_0, \xi_1, \cdots, \xi_m)$ is a p-basis for K over L, then by the remarks of § 1, there is a field k' with $K \supset L \supset k' \supset k$ such that $k'(\xi_i)$ is a simple extension of k' of exponent $n + 1$ for $i = 0, 1, \cdots, m$ and such that $K = \bigotimes_{i=0}^m k'(\xi_i)$, the tensor product being relative to k'. We may view K as an extension of k' and therefore, without loss of generality, may assume that $k' = k$. Let $\xi_i^{p^{n+1}} = c_i \in k$, $(x_0, \cdots, x_m)$ be commuting variables, and I be the ideal of $k[x_0, \cdots, x_m]$ generated by $x_i^{p^{n+1}} - c_i$, $i = 0, 1, \cdots, m$. Recalling that we have denoted $\mathbf{Z}[x_0, \cdots, x_m]$ by R_m in § 3, we have $k[x_0, \cdots, x_m] = k \otimes_{\mathbf{Z}} R_m$, whence K is isomorphic to $(K \otimes R_m)/I$.

Now if V is a vector space of dimension N over a field L, and φ a nilpotent endomorphism of index N, then the necessary and sufficient condition that an element v of V be of the form φw is that $\varphi^{N-1}v = 0$. We may therefore choose the p-basis (ξ) as follows. For ξ_0 take any element of K such that $\varphi\xi_0 = 1$. Having chosen $\xi_0, \xi_1, \cdots, \xi_{i-1}$, choose ξ_i such that $\varphi\xi_i = g_i(\xi_0, \xi_1, \cdots, \xi_{i-1})$, where g_i is the polynomial with integer coefficients defined in the preceding section by setting $\gamma_0 x_i = g_i(x_0, x_1, \cdots, x_{i-1})$. This is possible since φ^{p^i} annihilates $\xi_0, \xi_1, \cdots, \xi_{i-1}$ and therefore annihilates $g_i(\xi_0, \xi_1, \cdots, \xi_{i-1})$. That (ξ) is a p-basis then follows from the preceding lemma.

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The derivation $m\gamma_0$ of R_m induces a derivation $k \otimes m\gamma_0$ of $k \otimes R_m$. The latter derivation annihilates I and therefore induces a derivation of $k \otimes R_m/I = K$. The p-basis has been so chosen that this derivation is just φ. It is easy to see that, if we set $t^N = 0$, where $N = p^{n+1}$, then $k \otimes \Gamma_t$ is an abelian element of $\mathrm{Aut}_{N-1} K$ beginning $1 + t\varphi + \cdots$. Therefore $\varphi \in \mathrm{Der}_n^a K$, as asserted.

5. Deformation of a quotient algebra

Let $\Pi_t = 1 \frown 1 + tF_1 + t^2F_2 + \cdots$ be a one-parameter family of deformations of an algebra A over a ring k. If A contains an ideal I with the property that $F_i(x, a), F_i(a, x) \in I$ for all $x \in I$, $a \in A$ and all $i = 1, 2, \cdots$, then Π_t induces a one-parameter family of deformations of A/I. We shall say that I is stable under Π_t. If A is a $\mathbf{Q}$-algebra, i.e., if there is an unital morphism $\mathbf{Q} \to k$, and if $\Pi_t = \exp t\varphi \frown \psi$ where φ, ψ are commuting derivations of A, then a sufficient condition for the stability of I is that $\varphi I, \psi I \subset I$. Suppose that A is $\mathbf{Z}_p$ algebra and that $(\varphi) = (\varphi_0, \varphi_1, \cdots)$, $(\xi) = (\psi_0, \psi_1, \cdots)$ are commuting elements of $C^1(A, A)$ such that $e(t, (\varphi))$, $e(t, (\psi))$ are one-parameter families of automorphisms of A. Then a sufficient condition for the stability of an ideal I under $e(t, (\varphi), (\psi))$ is that $\varphi_i I, \psi_i I \subset I$ for all i.

6. Construction of a class of division algebras

Let $(x) = (x_0, x_1, \cdots)$ be a sequence of commuting variables, and, as before, write $R = \mathbf{Z}[x] = \mathbf{Z}[x_0, x_1, \cdots]$, $R_n = \mathbf{Z}[x_0, x_1, \cdots, x_n]$. If k is any commutative associative ring with unit, then $k \otimes_{\mathbf{Z}} R = k[x_0, x_1, \cdots]$. The one-parameter family Γ_t of automorphisms of R introduced in § 3 and defined by setting $w_i(\Gamma_t x) = w_i(x) + t^{p^i}$ induces a one-parameter family $k \otimes \Gamma_t$ of automorphisms of $k[x]$. By abuse of notation, we shall frequently make no distinction between the element x_i of R and its image in $k[x]$, and we shall denote $k \otimes \Gamma_t$ simply by Γ_t. Similarly, $\Pi_t = \exp \sum_{i=0}^{\infty} (t^{p^i}/p^i) w_i(\gamma) \frown w_{i+1}(\gamma)$ induces a one-parameter family of deformations of $k[x]$ which will be denoted, still, by Π_t. Then $k \otimes R_n[t] = k[x_0, \cdots, x_n, t]$ is stable under Γ_t and Π_t, consequently also induces a one-parameter family of deformations of $k[x_0, \cdots, x_n]$. The restrictions of Γ_t, γ_i, and Π_t to $k[x_0, \cdots, x_n]$ will be denoted by ${}_n\Gamma_t$, ${}_n\gamma_i$, and ${}_n\Pi_t$, respectively, whenever a distinction is necessary.

We suppose henceforth that R is a field of characteristic p. The following is then an immediate corollary of Theorem 18.

LEMMA 9. *Let elements $c_0, c_1, \cdots, c_n$ of k be given and I be the ideal of $k[x_0, \cdots, x_n]$ generated by $x_0^{p^{n+1}} - c_0, \cdots, x_n^{p^{n+1}} - c_n$. Then $\gamma_i I \subset I$ for $i = 0, 1, \cdots, n$.*

Now observe that ${}_n\Pi_t$ is a polynomial in t with coefficients which are cup

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products of polynomials in ${}_n\gamma_0, {}_n\gamma_1, \cdots, {}_n\gamma_n$. Therefore, by the remarks of the preceding section, Π_t induces a one-parameter family of deformations of $k[x_0, \cdots, x_n]/I$. If $c_0, \cdots, c_n$ are so chosen that $K = k[x_0, \cdots, x_n]/I$ is again a field, then by the remarks in § 2, $K(t)$ with the ${}_n\Pi_t$ multiplication is a division algebra, $D(c_0, \cdots, c_n)$, of dimension $p^{(n+1)^2}$ over $k(t)$. The center $\mathfrak{Z}(c_0, \cdots, c_n)$, will be shown to be a simple, purely inseparable extension of exponent $n+1$ over $k(t)$. The main open question is whether $D(c_0, \cdots, c_n)$, which has dimension $p^{(n+1)^2-(n+1)}$ over its center, is a crossed product. It is known that a finite dimensional central division algebra over an algebraic number field or a field of transcendence degree one over a finite field is in fact cyclic. The nature of the present construction, however, is such that the center of $D(c_0, \cdots, c_n)$ is of transcendence degree at least two. For the center contains $k(t)$, k contains at least one transcendental else it would have no finite inseparable extensions, and the "deformation parameter" t contributes a second. It follows from elementary results of Teichmüller and Albert [15], [1], that $D(c_0, c_1)$ is a crossed product over $\mathfrak{Z}(c_0, c_1)$, for the dimension is then just p^2. It seems reasonable to conjecture, however, that if $n \geqq 2$, then $D(c_0, \cdots, c_n)$ is in general not a crossed product.

In the determination of $\mathfrak{Z}(c_0, \cdots, c_n)$ we shall need the following remarks which are special cases of the results of Chapter II, § 5.

1. γ_0 is a derivation of $K = k[x_0, \cdots, x_n]/I$, the restriction of γ_1 to the null space of γ_0 is a derivation on that null space, and in general, if $K_i = \{y \in K \mid \gamma_0 y = \gamma_1 y = \cdots = \gamma_i y = 0\}$, then the restriction of γ_{i+1} to K_i is a derivation.

2. γ_0 is a derivation of $D = D(c_0, \cdots, c_n)$, i.e., γ_0 is still a derivation with respect to the Π_t multiplication, and if we set $D_i = \{y \in D \mid \gamma_0 y = \cdots = \gamma_i y = 0\}$, then the restriction of γ_{i+1} to D_i is a derivation of D_i.

The image of x_i in $k[x_0, \cdots, x_n]/I$ in K will be denoted by ξ_i; one has $\xi_i^{p^{n+1}} = c_i \in k$. Henceforth, γ_i will tacitly denote ${}_n\gamma_i$, and similarly for Γ_t and Π_t.

LEMMA 10. *Let* $K_i = \{y \in K \mid \gamma_0 y = \gamma_1 y = \cdots = \gamma_i y = 0\}$. *Then* $K_i = k(\xi_0^{p^{i+1}}, \cdots, \xi_n^{p^{i+1}})$ *for* $i = 0, 1, \cdots, n$. *In particular,* $K_n = k$.

PROOF. Observe first that γ_0 is a nilpotent derivation of index p^{n+1}, for by Equation (3.3), $\gamma_0^{p^{n+1}} = 0$, while by Lemma 7, $\gamma_0^{p^n} \neq 0$. It follows from Lemma 8 that $[K : K_0] = p^{n+1}$. Since $\xi_0^p, \cdots, \xi_n^p \in K_0$, and $[K : k(\xi_0^p, \cdots, \xi_n^p)] = p^{n+1}$, it follows that $K_0 = k(\xi_0^p, \cdots, \xi_n^p)$. This proves the lemma for $i = 0$. Suppose that the lemma is true for some $i \geqq 0$. Then the restriction of γ_{i+1} to $k(\xi_0^{p^{i+1}}, \cdots, \xi_n^{p^{i+1}})$ is a derivation. By Theorem 16 we have $\gamma_{i+1} a^{p^{i+1}} = (\gamma_0 a)^{p^{i+1}}$

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for all $a \in K$. It follows that the restriction of γ_{i+1} to $k(\xi_0^{p^{i+1}}, \cdots, \xi_n^{p^{i+1}})$ is nilpotent of index p^{n+1}, so by the same argument as for the case $i = 0$, one has $K_{i+1} = k(\xi_0^{p^{i+2}}, \cdots, \xi_n^{p^{i+2}})$. This completes the induction and the proof.

The parameters $c_0, c_1, \cdots, c_n$ being henceforth understood, $D(c_0, \cdots, c_n)$ will be denoted simply by D and its center by z. Set $q = p^{n+1}$. If we consider Γ_t modulo t^q, then it is an element of $\mathrm{Aut}_{q-1} K$. It follows by the remarks of Chapter II, § 4 that Γ_u is an element of $\mathrm{Aut}_{q-1} D$. (The parameter has been changed to u avoid confusion, since t is an element of D.) Setting

$$D_i = \{a \in D \mid \gamma_0 a = \gamma_1 a = \cdots = \gamma_i a = 0\} ,$$

the underlying $k(t)$-vector space of D_i is $k(t) \otimes_k K_i$, and the latter is therefore a subalgebra of D in the Π_t multiplication.

LEMMA 11. *If $a \in D_{i-1}$, then $\Pi_t(\xi_i, a) = \xi_i a + t^{p^i} \gamma_i^p a$, and $\Pi_t(a, \xi_i) = \xi_i a$.*

PROOF. Consider $p^{-j}\langle w_j(\gamma), \xi_i\rangle\langle w_{j+1}(\gamma), a\rangle$, an expression which is meaningful even though the characteristic is p. The second factor vanishes if $j < i$, and the first vanishes if $j > i$. One has $p^{-i}\langle w_i(\gamma), \xi_i\rangle\langle w_{i+1}(\gamma), a\rangle = \gamma_i^p a$, whence $\Pi_t(\xi_i, a) = \xi_i a + t^{p^i}(\gamma_i^p a)$. On the other hand, $\langle w_j(\gamma), a\rangle\langle w_{j+1}(\gamma), \xi_i\rangle$ vanishes for all i, so $\Pi_t(a, \xi_i) = \xi_i a$.

COROLLARY. *If $z \in \mathfrak{Z} \cap D_{i-1}$, then $\gamma_i^p z = 0$.*

Let A be any algebra over a ring k, $\mathfrak{Z}$ be the center of A, and $\Phi_u = 1 + u\varphi_1 + \cdots + u^m \varphi_m$ be an element of $\mathrm{Aut}_m A$ for some m. Then $\varphi_i(\mathfrak{Z}) \subset \mathfrak{Z}$, $i = 1, \cdots, m$. For $\Phi_u b \cdot \Phi_u z \equiv \Phi_u(bz) \equiv \Phi_u(zb) \equiv \Phi_u z \cdot \Phi_u b \bmod t^{m+1}$ for all $b \in A[[u]]$ and all $z \in \mathfrak{Z}$. Setting $b = \Phi_u^{-1} a$, we have $a(\Phi_u z) \equiv (\Phi_u z)a$ for all $a \in A$, which proves the assertion. In particular, it follows that, if we write $\Gamma_u = 1 + u\varphi_1 + u^2\varphi_2 + \cdots + u^{q-1}\varphi_{q-1}$, when $q = p^{n+1}$, then $\varphi_i \mathfrak{Z} \subset \mathfrak{Z}$. Since by Theorem 20, γ_i is a polynomial in $\varphi_1, \cdots, \varphi_{p^i}$, it follows that $\gamma_i \mathfrak{Z} \subset \mathfrak{Z}$ for $i = 0, 1, \cdots, n$.

THEOREM 22. *The center $\mathfrak{Z}$ of D is a simple purely inseparable extension of exponent $n + 1$ of $k(t)$.*

PROOF. Set $\mathfrak{Z}_{i-1} = \mathfrak{Z} \cap D_{i-1}$. Since γ_i is a derivation of D_{i-1} and carries $\mathfrak{Z}$ into itself, it follows that γ_i is a derivation of the field $\mathfrak{Z}_{i-1}$. Moreover, its null space is precisely $\mathfrak{Z}_i$. From the preceding lemma and from Lemma 8, it follows that $[\mathfrak{Z}_{i-1}: \mathfrak{Z}_i] \leqq p$. Since

$$[\mathfrak{Z}: k(t)] = [\mathfrak{Z}: \mathfrak{Z}_0][\mathfrak{Z}_0: \mathfrak{Z}_1] \cdots [\mathfrak{Z}_{n-1}: \mathfrak{Z}_n (= k(t))] ,$$

it follows that $[\mathfrak{Z}: k(t)] \leqq p^{n+1}$. We show next that the restriction of γ_0 to $\mathfrak{Z}$ is not zero. For suppose $\gamma_0(\mathfrak{Z}) = 0$. Then γ_0 is a derivation of D over $\mathfrak{Z}$ and hence is inner, i.e., $\gamma_0 = \mathrm{ad}\, y$ for some $y \in D$. Since $(\mathrm{ad}\, y)y = 0$, we have

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$\gamma_0 y = 0$, whence $y \in D_0$. Let $y^{\cdot p}$ denote the pth power of y *in the multiplication in* D. Then $y^{\cdot p} \in D_0$, and $\operatorname{ad}(y^{\cdot p}) = (\operatorname{ad} y)^p = \gamma_0^p$. But from the preceding lemma we have $\operatorname{ad} \xi_0 = \gamma_0^p$, whence $\xi_0 - y^{\cdot p} \in \mathfrak{Z}$. Since $\gamma_0(\mathfrak{Z}) = 0$, this implies that $\gamma_0(\xi_0) = 0$, contradicting the fact that $\gamma_0(\xi_0) = 1$. Therefore, $\gamma_0(\mathfrak{Z}) \neq 0$.

Now $\Gamma_u \in \operatorname{Aut}_{q-1} D$, where $q = p^{n+1}$, so $\gamma_0 \in \operatorname{Der}_n D$. Since Γ_0 carries $\mathfrak{Z}[u]$ into itself, it follows that $\operatorname{Der}_n \mathfrak{Z} \neq 0$, for it contains the restriction of γ_0 to $\mathfrak{Z}$. Thus $\mathfrak{Z}$ has exponent at least $n + 1$ over its ground field $k(t)$. Since $[\mathfrak{Z}: k(t)] \leqq p^{n+1}$, the only possibility is that the extension be simple and of exponent precisely $n + 1$.

Open problems at present include

(1) giving an explicit generator for $\mathfrak{Z}$;

(2) determining when $D(c_0, \cdots, c_n)$ and $D(c_0', \cdots, c_n')$ are isomorphic; and of course

(3) determining which of the algebras $D(c_0, \cdots, c_n)$ are crossed products.

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